INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 4: Implementation and Empirical Results

by

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4 Implementation and Empirical Results

This chapter runs the design of Chapter 3 on Arbitrum One as a single case study: one chain and one score window per freeze, one matched cohort of contracts for every same-window comparison, and spender cohorts for the two label windows W0 and W1. It reports the contract-graph analysis, the main analysis of the thesis (Section 3.5.3): the data (Section 4.1); the cost, agreement, divergence and robustness of the two single-layer walks (Sections 4.2 to 4.5); the discovery-order test (Section 4.6); the out-of-window prediction of the single-layer walks against the counts they smooth (Section 4.7); and C-PR, the method of the thesis, under its rules and beside every other score (Section 4.8). Rules A and B were fixed in the analysis plan and in the registration of W1 for the raw-unit analysis; here they are applied to the scores of the contract graph as a re-analysis, and the registered analysis gave the same readings (Appendix 8.6, Table 8.16). Each Kendall τ comes with its confidence interval except in the value-slope and sample-definition checks, and each walk is read against the raw count it smooths.

4.1 Case study setting: Arbitrum One

Arbitrum One is a heavily used Ethereum rollup with mature DeFi protocols (Section 2.2), and its logs are in the public BigQuery export that the pipeline reads. The score windows run from 1 July 2025 to 30 June 2026 for the same-window evaluation and W1 and from 1 April 2025 to 31 March 2026 for W0, and the time decay is anchored at the freeze (Figure 3.3). After the registered results were known, the revised analysis was defined: amounts in US dollars on five tokens; scores from the twelve months before each freeze; graphs whose nodes are contracts other than smart accounts, with edges read from the whole chain; each allowance counted up to its owner's balance at the freeze, with balances reconstructed from `Transfer` logs; constants set on the con-

tract graph; and the discovery-order test. That analysis is the one reported here, so no result of this chapter is a registered result.

From 1 April 2025 to 30 September 2026 the five tokens (WETH, USDC, USDT0, ARB and WBTC) have 1,240,275,042 `Approval` and non-zero `Transfer` rows, which name 25,537,280 addresses other than the zero address. The 17,477,779 that sent a transaction are user accounts. The addresses that could still end an edge between contracts, 6,230,270 with the cohort members and two audit samples, were read with `eth_getCode` at block 513,216,506; those without code, EIP-7702 accounts, Safes and ERC-4337 accounts are user accounts too, which leaves 1,829,719 contracts (Section 3.5.1; Table 8.4 in Appendix 8.4.3). Only the 603,475,784 rows with a contract at both ends can make an edge: no user account is a node, and no score or baseline reads an edge of one. Figure 4.1 traces these counts from the rows to the contract graph.



Figure 4.1: From the rows of the five tokens to the contract graph. The upper lane counts the `Approval` and non-zero `Transfer` rows of WETH, USDC, USDT0, ARB and WBTC from 1 April 2025 to 30 September 2026, the lower lane the addresses they name; bar lengths are proportional to the counts within a lane. Transaction senders and the addresses that their code shows to be user accounts leave the funnel (dashed), and only the rows with a contract at both ends can make an edge.

The contract graph numbers 1,403,959 contracts over both freezes and has two layers (Table 4.1): transfer pairs join a sender contract to a recipient contract, and allowance pairs an owner contract to a spender contract whose latest allowance of the window is positive and stays positive under the balance cap (Section 3.6). The cap decides the size of that layer. At T_{obs}, 223,892 of the 339,282

positive latest allowances between contracts are unlimited (66.0%), 222,559 are capped to zero because the owner holds none of the token (65.6%), and 116,723 stay positive; at t_1, 289,299 of 308,922 are capped to zero (93.6%) and 19,623 stay positive (Table 8.5 in Appendix 8.4.3). For 1,493 of 1,500 sampled (token, owner) pairs (99.5%) the reconstructed balance (Section 3.5.2) equals `balanceOf` at the last block of 30 June 2026.

Table 4.1: The two layers of the contract graph at both freezes: ordered pairs of contracts, the distinct contracts at their source (owners or senders) and destination (spenders or recipients), and the largest number of pairs that one source and one destination hold. Every end of every pair is a contract; the graph numbers 1,403,959 contracts over both freezes. At T_{obs} one owner contract holds 98,073 of the 115,006 capped allowance pairs; at t_1 the largest owner holds 256 of 17,940.

Layer	Pairs	Sources	Destinations	Largest source	Largest destination
At T_{obs}					
Allowance layer (capped)	115,006	9,459	100,112	98,073	2,097
Allowance pairs before the cap	302,559	141,974	108,825	98,073	28,082
Transfer layer	2,580,709	1,007,310	801,813	130,667	441,612
At t_1					
Allowance layer (capped)	17,940	10,153	2,050	256	2,008
Allowance pairs before the cap	275,917	140,327	89,727	78,382	25,810
Transfer layer	2,269,073	873,388	626,356	121,708	439,299

The allowance layer is far smaller than the transfer layer: 115,006 pairs against 2,580,709 at T_{obs} (4.5%) and 17,940 against 2,269,073 at t_1. At T_{obs} it is also concentrated: one owner contract, which this study did not identify, holds 98,073 of the 115,006 pairs, whereas the largest owner at t_1 holds 256. The allowance-side results at T_{obs}, those of the same window and of W1, are read with this in mind, and W0 gives the same comparisons on a layer without such an owner. The union graph, the distinct pairs of different contracts with an edge in either layer, has 2,575,632 pairs over 1,299,438 contracts at T_{obs} and 2,269,421 over 1,111,554 at t_1. The constants, set on the W0 score window of the contract graph, are those of Section 3.5.3: $m_T = 160$ and $m_A = 0.071$ dollars, $U = 16{,}000$ dollars for C-PR's allowance layer, and $U_{\text{user}} = 31{,}000$ dollars for the importance measures of Section 3.7.

4.1.1 Cohorts

The rules of the three contract sets below count every approving account, user accounts included, so membership does not depend on the contract graph; Table 4.2 shows where the members lie in it.

- Matched cohort ($n = 6{,}443$). The rule of Section 3.2, a non-zero `Approval` of one of the five tokens from at least three distinct owners in the score window of T_{obs}, lists 6,483 addresses with code; 40 of them are smart accounts and so user accounts, and the cohort is the other 6,443. Every same-window alignment, difference test and benchmark is computed on it. The same rule in the W0 score window lists 6,343 addresses, of which $6{,}292$ are contracts; the spenders and labels of each window are drawn from the events around the addresses listed for its freeze.
- Spender cohorts of W0 and W1 ($n = 7{,}864$ and $7{,}915$). In W0 the scores are frozen at t_1 (31 March 2026, 23:59:59 UTC), and the labels are counted for every spender that holds, in the events around the addresses listed for W0, a positive latest allowance of one of the five tokens at t_1, set inside the W0 score window. The cohort is the 7,864 contracts among these 7,962 spenders; 97 are user accounts and 1 is a system address. In W1 the scores are frozen at T_{obs}, and the same rule gives 7,915 contracts among 7,995 spenders. A spender needs only one owner, so these cohorts are not subsets of the matched cohort.
- Subsamples of the matched cohort ($n = 1{,}000$, $2{,}000$ and $5{,}000$). Deterministic SHA256 subsamples (seed `contract-benchmark-v1`), used only for runtime against size and for the sensitivity to sample definition (Section 3.13.1). Each stage keeps the pairs incident to its sampled contracts.

The contract graph does not reach every member. Of the matched cohort, 1,250 contracts (19.4%) end a capped allowance pair, 4,059 (63.0%) a transfer pair and 4,270 (66.3%) a pair of the union graph; the other 2,173 have no transfer with another contract in the window and no allowance to or from one that stays positive under the cap. Of the W0 spenders 5,700 (72.5%) and of the W1

Table 4.2: Where the members of each cohort lie in the contract graph of their freeze: members that end a capped allowance pair, a transfer pair, a pair of the union graph, and members that end no pair between contracts. A member outside a graph stays in its cohort and scores zero under every score and baseline, so the scores tie on these members; Table 4.23 repeats the main comparisons on the members inside the union graph.

Cohort	Members	In the allowance layer	In the transfer layer	In the union graph	Outside the graph
Matched cohort at T_{obs}	6,443	1,250 (19.4%)	4,059 (63.0%)	4,270 (66.3%)	2,173 (33.7%)
W0 cohort at t_1	6,292	1,248 (19.8%)	3,893 (61.9%)	4,108 (65.3%)	2,184 (34.7%)
W0 spenders at t_1	7,864	1,901 (24.2%)	5,443 (69.2%)	5,700 (72.5%)	2,164 (27.5%)
W1 spenders at T_{obs}	7,915	1,975 (25.0%)	5,504 (69.5%)	5,759 (72.8%)	2,156 (27.2%)

spenders 5,759 (72.8%) lie in the union graph. A member outside a graph stays in its cohort and scores zero under every score and baseline of that graph, so the scores tie on these members. Every coefficient of this chapter therefore reflects both how a score orders the members that the graph reaches and how it sets them apart from the others; Table 4.23 repeats the main out-of-window contrasts inside the union graph.

Contracts that one actor deploys can send transfers or grant allowances only among themselves, and a walk keeps the mass that reaches a group with no edge out of it (Langville & Meyer, 2006); link farms raise PageRank in this way (Gyöngyi & Garcia-Molina, 2005). A check made after every result was known, which decides no rule, therefore looked for such closed groups in the graph of each walk: sets of two or more contracts that are strongly connected and have no edge to a contract outside the set. The allowance layer has none at either freeze. The transfer layer has 6,860 at T_{obs}, with 31,563 contracts (2.4% of its contracts) and 4.9% of AWP's stationary mass, and 492 at t_1, with 2,846 contracts and 0.3% of that mass; in the graph of C-PR they hold 6.0% and 0.7% of its mass. They contain 72 of the 6,443 contracts of the matched cohort and 82 of the 6,292 of the W0 cohort, and at most one of the hundredth that any walk ranks highest in a cohort or spender set. Such groups therefore do not shape the rankings of the contracts studied here; the check does not show who controls a contract.

The solver graphs of EndorseRank and AWP in Table 4.3 do not hold the 9 allowance pairs and 328 transfer pairs whose value weight is zero in floating point, nor the 8 and 75 contracts that end

no other pair; the layers of C-PR, whose weights are sums of amounts, hold them. Table 8.3 in Appendix 8.4.2 gives the five tokens with their allowances and price checks.

4.1.2 Implementation

Every algorithm runs on one Python implementation of PageRank ($d = 0.85$, $\varepsilon = 10^{-8}$, $I_{\max} = 300$); the methods differ in their edge lists and restart vectors. The tables of this chapter come from the summary files of the evaluation runs, the benchmark and the discovery-order test, as collected in the results digest. At $d = 0.85$ every solve of the same-window and out-of-window evaluations converged well inside the iteration cap. Appendix 8.1 lists the runs, the repository revision, the configuration file and the summary files.

4.2 Computational benchmark (Claim C6)

Claim C6 concerns what each score costs to compute on the same input. Table 4.3 reports the cost of each algorithm on the contract graph at T_{obs}, on which the matched cohort is scored, following the timing procedure of Section 3.12. Each graph and setting was timed once, so the same measurement reappears as the last row of Table 4.4 and as the $d = 0.85$ row of Table 4.7; Table 8.1 in Appendix 8.1.4 lists every timed run.

EndorseRank solves the allowance graph ($114{,}997$ edges) in 0.101 s and AWP the transfer graph ($2{,}580{,}381$ edges) in 2.82 s, so AWP takes 27.9 times as long, at an edge ratio of 22.4. The cost of an iteration grows with $|E|$ and with $|V|$ (Appendix 8.3.2). Per iteration, assembly included, AWP's solve takes 15.9 times as long as EndorseRank's (0.029 against 0.0018 s), between the node ratio of 11.9 and the edge ratio, and AWP needs 1.75 times as many iterations (98 against 56); together the two factors give the time ratio (for other d, Section 4.5.1). Both scores use the same solver, so the difference in speed comes from the size of each graph and from the iterations its walk needs there, which the benchmark does not separate; it is no advantage of the EndorseRank algorithm.

C-PR walks on the union of both layers ($2{,}591{,}253$ distinct pairs, the pairs of a contract with itself

Table 4.3: Computational benchmark on the contract graph at T_{obs}, which holds every pair between contracts and not only the pairs of the matched cohort (AWP/EndorseRank runtime ratio 27.9× at an edge ratio of 22.4×). Runtime = mean wall-clock seconds over 5 timed solver calls after one untimed warm-up, on which alone peak memory is traced; SD over the same 5 runs. A timed call turns an edge list into scores: node indexing, sparse-matrix assembly, row normalization, restart vector and power iteration. $|V|$ and $|E|$ are the nodes and distinct edges of each method's solver graph. C-PR walks on the union of the allowance layer and the transfer layer, so its $|E|$ counts the distinct pairs of both; S-PR runs on the transfer layer, and its runtime and iterations include the solve of the allowance layer that supplies its restart vector. Table 8.1 lists every run.

Method	Runtime (s)	SD (s)	Peak mem. (MB)	Iters	$\lvert V\rvert$	$\lvert E\rvert$
EndorseRank	0.101	0.009	13.7	56	109,381	114,997
AWP	2.821	0.189	228.2	98	1,298,649	2,580,381
C-PR ($\lambda = 0.5$)	3.080	0.184	204.8	92	1,299,479	2,591,253
S-PR	2.657	0.202	142.6	153	1,298,724	2,580,709

included). Its solver keeps one matrix per layer and multiplies both in every iteration; it converges in 92 iterations and takes 3.08 s, 1.09 times as long as AWP. S-PR takes 2.66 s and 153 iterations, the solve of the allowance layer that supplies its restart vector included. Every algorithm that walks the transfer layer, alone or with allowance edges, takes time in proportion to that layer's size.

The time covers only the solver call on edge lists built in advance, not the reading of events, the typing of accounts, the balances or the edge lists, which all algorithms share wholly or in part.

4.2.1 Runtime on subsamples of the matched cohort

Table 4.4 times EndorseRank and AWP on three subsamples of the matched cohort and, in its last row, on the whole contract graph.

Few of the pairs that a subsample keeps are allowance pairs: the transfer graph has 74.8, 49.3 and 79.2 times as many edges as the allowance graph in the three subsamples. If the time of a solve followed $|E|$ alone, the time ratio would follow these edge ratios; it is 74.2, 60.2 and 108.0, that is 0.99, 1.22 and 1.36 times the edge ratio, with 48, 26 and 62 iterations for EndorseRank against 94, 90 and 104 for AWP. The iterations each walk needs on each subgraph, and not the edges alone, set the ratio. The last row, the measurement of Table 4.3, holds every pair between contracts and not

Table 4.4: Runtime of EndorseRank (ER) and AWP on subsamples of the matched cohort and on the whole contract graph. The numbered rows are deterministic SHA256 subsamples of the cohort (seed `contract-benchmark-v1`); each keeps the pairs incident to its n sampled contracts, so $|V|$ and $|E|$ of both solver graphs are reported per row. AWP/ER time and AWP/ER $|E|$ are the ratios of the two runtimes and of the two edge counts in the same row. Runtime = mean wall-clock seconds over 5 timed solver calls after one untimed warm-up. [a]The measurement of Table 4.3 on the whole contract graph, which holds every pair between contracts; 16,324 allowance pairs and 1,063,593 transfer pairs are incident to the full cohort of $n = 6{,}443$ contracts, and the graph of those pairs has no timing of its own.

n	ER (s)	AWP (s)	AWP/ER time	AWP/ER $\|E\|$	ER $\|V\|$	ER $\|E\|$	AWP $\|V\|$	AWP $\|E\|$
1,000	0.001	0.095	74.2×	74.8×	1,293	1,608	78,869	120,252
2,000	0.002	0.134	60.2×	49.3×	3,972	4,784	129,533	235,864
5,000	0.006	0.689	108.0×	79.2×	8,720	11,240	426,557	889,662
Whole graph[a]	0.101	2.821	27.9×	22.4×	109,381	114,997	1,298,649	2,580,381

only the pairs incident to the cohort, so the step from the third row to the last is a change of graph more than of n. The rows show how runtime grows with the edges up to the size of the whole graph and not beyond it (Figure 4.2).

4.3 Same-window alignment (Claim C5)

Claim C5 asks what each score agrees with in the window it is computed from. Table 4.5 reports the family-mean Kendall τ of every score on the transfer, allowance and stability families; the proxies are computed from the events between contracts (Section 3.10). Apart from the secondary part of rule A, fixed in the analysis plan (Section 3.8) and applied here as a re-analysis, the comparisons below belong to no rule.

The scores divide by the edges they walk on. The two that walk on transfers alone, AWP and C-PR ($\lambda = 0$), reach 0.755 ($[0.748, 0.762]$) and 0.804 ($[0.799, 0.809]$) on the transfer family and 0.227 ($[0.206, 0.245]$) and 0.238 ($[0.218, 0.258]$) on the allowance family. The two that walk on allowances alone, EndorseRank and C-PR ($\lambda = 1$), reach 0.839 ($[0.828, 0.849]$) and 0.838 ($[0.827, 0.850]$) on the allowance family and 0.297 ($[0.280, 0.317]$) and 0.298 ($[0.281, 0.318]$) on the transfer family; S-PR and the interior mixtures of C-PR lie between the two layers. On the

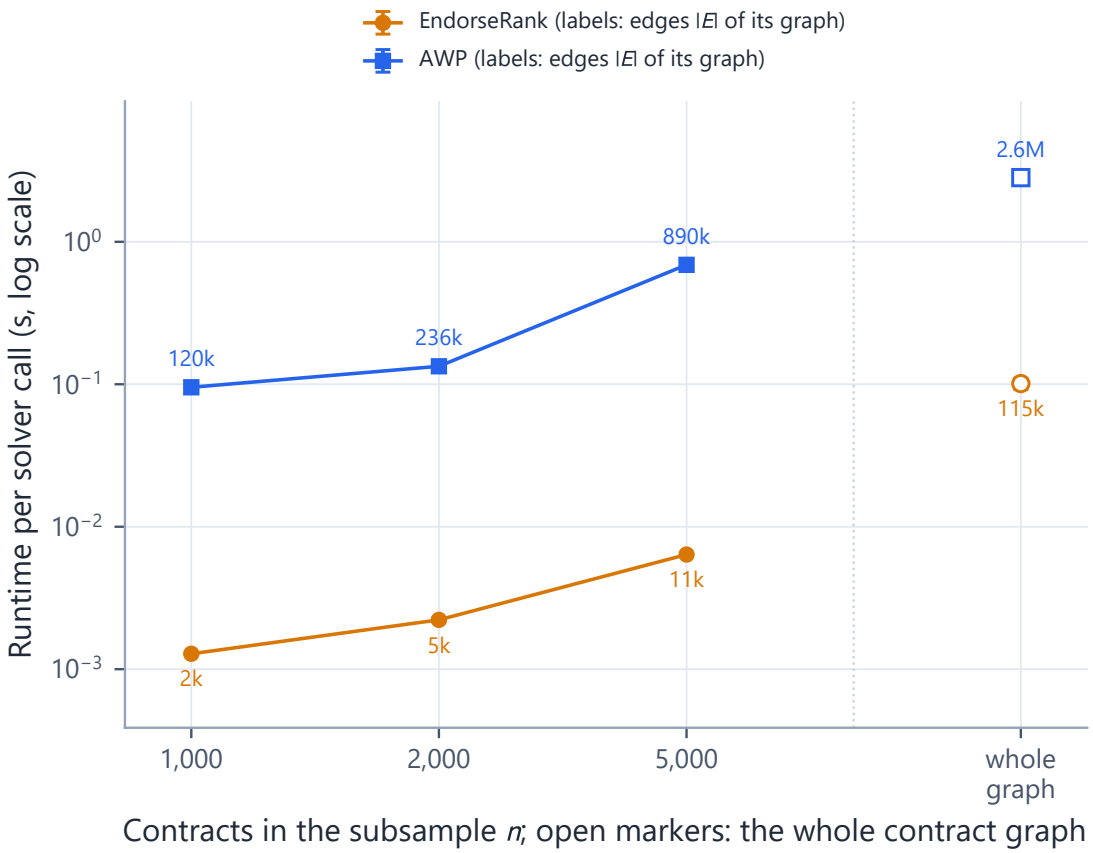


Figure 4.2: Wall-clock time of one solver call of EndorseRank and AWP for each row of Table 4.4, on a log scale: the three subsamples of the matched cohort (filled markers, joined by lines) and the whole contract graph (open markers). The label beside each point gives the edge count $|E|$ of that graph, and error bars are the standard deviations over five timed runs.

stability family the scores that walk on transfers lead, AWP with 0.643 ($[0.635, 0.651]$), while EndorseRank stays at 0.220 ($[0.206, 0.237]$). Part of this agreement is built in: the main proxies of each family come from the same events as one of the layers, in-degree and in-value from the transfers and in-approve degree and value from the capped latest allowances, and a PageRank smooths the in-degree of its own graph. Agreement with the proxies of a score's own edge type is therefore an intended-construct check, not an external test.

On the allowance side the reach of the layer builds in a second part. EndorseRank sets only the 926 cohort contracts with an in-edge in its graph above the rest: it gives 317 others, which are

Table 4.5: Same-window family-mean Kendall τ with bootstrap intervals for every score of this study (matched cohort, $n = 6{,}443$ contracts). The scores are EndorseRank and AWP, the coupled operator C-PR at five values of λ and the endorsement-seeded walk S-PR; $\lambda = 1$ is the allowance layer walked alone (latest allowances between contracts in USD, each counted up to its owner's balance and capped) and $\lambda = 0$ the transfer layer walked alone (USD amounts discounted by the decay of AWP); both restart uniformly. The raw degrees have no row because they are proxies of the transfer and allowance families; EndorseRank with activity restarts is in Table 4.10. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Method	Transfer	Allowance	Stability
EndorseRank	0.297 [0.280, 0.317]	0.839 [0.828, 0.849]	0.220 [0.206, 0.237]
AWP	0.755 [0.748, 0.762]	0.227 [0.206, 0.245]	0.643 [0.635, 0.651]
C-PR ($\lambda = 1$)	0.298 [0.281, 0.318]	0.838 [0.827, 0.850]	0.222 [0.207, 0.238]
C-PR ($\lambda = 0$)	0.804 [0.799, 0.809]	0.238 [0.218, 0.258]	0.637 [0.628, 0.646]
C-PR ($\lambda = 0.25$)	0.742 [0.734, 0.751]	0.424 [0.411, 0.437]	0.577 [0.567, 0.588]
C-PR ($\lambda = 0.5$)	0.738 [0.729, 0.747]	0.433 [0.419, 0.445]	0.574 [0.564, 0.585]
C-PR ($\lambda = 0.75$)	0.734 [0.725, 0.743]	0.438 [0.424, 0.450]	0.572 [0.561, 0.583]
S-PR	0.782 [0.773, 0.791]	0.371 [0.350, 0.391]	0.584 [0.575, 0.593]

in the graph only as owners, the score of a node without in-edges, and the remaining $5{,}200$ zero (Section 4.4). The in-approve degree and value of a contract that receives no capped allowance are zero. Most of the pairs that EndorseRank and these proxies order alike therefore set a contract that the layer reaches as a spender against one that it does not, and the coefficient of 0.839 says mainly that score and count agree on that boundary.

Appendix 8.5.2 breaks the family means of EndorseRank and AWP into their proxies. AWP follows the transfer in-degree (0.764, $[0.757, 0.772]$) and the in-value (0.746, $[0.738, 0.754]$) about equally, and EndorseRank the in-approve degree (0.847, $[0.835, 0.857]$) and the in-approve value (0.831, $[0.820, 0.841]$). C-PR ($\lambda = 1$) walks the allowance layer with capped amounts, also restarts uniformly, and stays within 0.002 of EndorseRank on every family.

The first half of Claim C5 is supported in a narrow sense: every score that walks on one edge type agrees most with the proxies of that edge type, the mixtures of C-PR lie between their two layers on every family, and the agreement is partly built in, by the shared events and by the reach of each layer. Section 4.8.1 locates C-PR within the window.

4.4 Rank divergence (H1 / RQ1)

Hypothesis H1 and RQ1 concern whether EndorseRank and AWP put the same contracts in different orders. The inter-method Kendall τ_b between the two score vectors is 0.275, with interval $[0.258, 0.294]$ (last row of Table 4.6), so the interval excludes both zero and one. Of the $20{,}752{,}903$ contract pairs, only 32.3% are ordered by both scores, and 26.7% of those pairs are discordant.

Ties explain the first of these shares (Figure 4.3). EndorseRank gives 801 distinct scores and ties 65.4% of the contract pairs; AWP gives $3{,}982$ and ties 13.7%. Under each score the largest tie block is formed by the contracts that score zero, $5{,}200$ under EndorseRank and $2{,}384$ under AWP: the $5{,}193$ and $2{,}384$ cohort contracts outside each layer and, under EndorseRank, 7 whose only pairs carry amounts too small for the value weight to differ from zero. Under EndorseRank 317 further contracts share the score $c = 8.45 \times 10^{-6}$ of a node without in-edges. In Figure 4.4, among the contracts that EndorseRank ranks above c, the points spread over most of AWP's range.

The zero blocks come from one rule of this pipeline: a contract missing from a graph scores zero (Section 3.6). The analysis plan fixed the alternative, keeping the missing contracts as isolated nodes, as a sensitivity analysis, and Table 4.6 gives it. Under uniform restarts an isolated node scores like any node without in-edges, so the $5{,}193$ contracts outside the allowance layer tie with the 317. EndorseRank's allowance coefficient then rises from 0.839 to 0.956 ($[0.952, 0.960]$), because the score now ties the contracts that the allowance proxies tie at zero, and its transfer coefficient falls from 0.297 to 0.259 ($[0.239, 0.278]$). The agreement between the two scores falls from 0.275 to 0.227 ($[0.207, 0.245]$): the zero rule sets the 317 above the $5{,}193$, and AWP on balance orders those pairs the same way. AWP restarts only at contracts that sent a transfer, so an isolated contract keeps the score zero and AWP's coefficients are the same under both rules.

H1 is supported: the two scores order the cohort differently, but the support is narrower than the coefficient suggests. Two thirds of the pairs are not ordered by both scores, because the allowance layer between contracts reaches a fifth of the cohort, and the rule for missing contracts, which leaves AWP's coefficients as they are, moves EndorseRank's. This divergence, the second half of

Table 4.6: Same-window family-mean Kendall τ with contracts missing from a graph scored zero or kept as isolated nodes (matched cohort, $n = 6{,}443$ contracts). The main analysis scores a contract that is missing from a method's graph zero; keeping it as an isolated node is the sensitivity analysis fixed in the analysis plan. The rule concerns 5,193 contracts for EndorseRank and 2,384 for AWP (2,173 are missing from both graphs). Under EndorseRank's uniform restarts an isolated node gets the score of a node without in-edges. AWP restarts only at contracts that sent a transfer, so an isolated contract keeps the score zero and AWP's coefficients are the same under both rules. The last row is the agreement between the two scores. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	Missing contracts scored zero		Missing contracts as isolated nodes	
Family	EndorseRank	AWP	EndorseRank	AWP
Transfer	0.297 [0.280, 0.317]	0.755 [0.748, 0.762]	0.259 [0.239, 0.278]	0.755 [0.748, 0.762]
Allowance	0.839 [0.828, 0.849]	0.227 [0.206, 0.245]	0.956 [0.952, 0.960]	0.227 [0.206, 0.245]
Stability	0.220 [0.206, 0.237]	0.643 [0.635, 0.651]	0.171 [0.155, 0.187]	0.643 [0.635, 0.651]
EndorseRank vs. AWP	0.275 [0.258, 0.294]		0.227 [0.207, 0.245]	

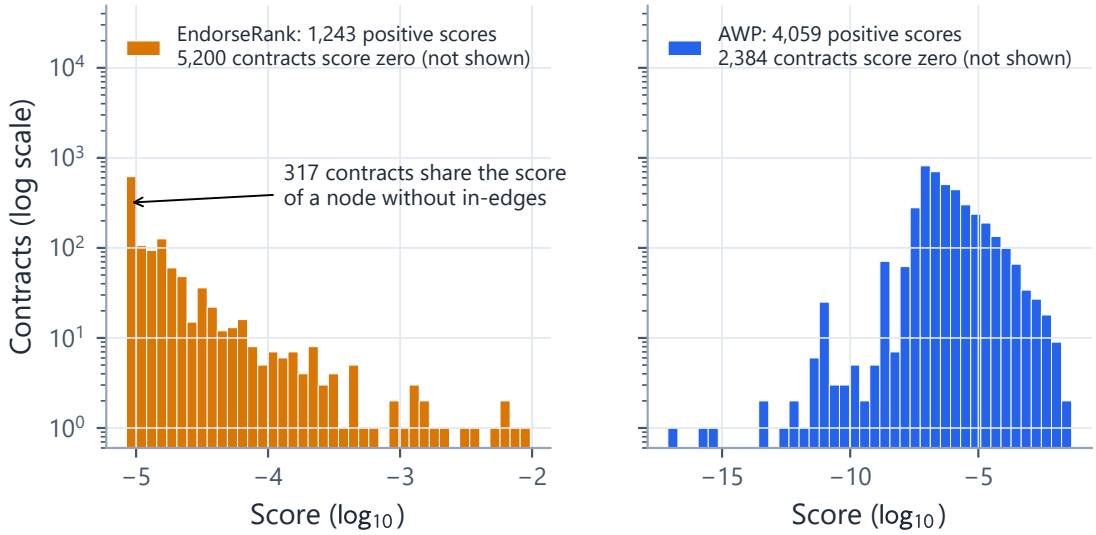


Figure 4.3: Marginal EndorseRank and AWP score distributions on the matched cohort ($n = 6{,}443$ contracts), drawn on a $\log_{10}$ scale for contracts with a positive score; counts are on a log scale. Zero-score contracts, which no edge of positive weight in the respective graph touches, are not drawn: 5,200 for EndorseRank and 2,384 for AWP. Under EndorseRank 317 contracts share the score of a node without in-edges.

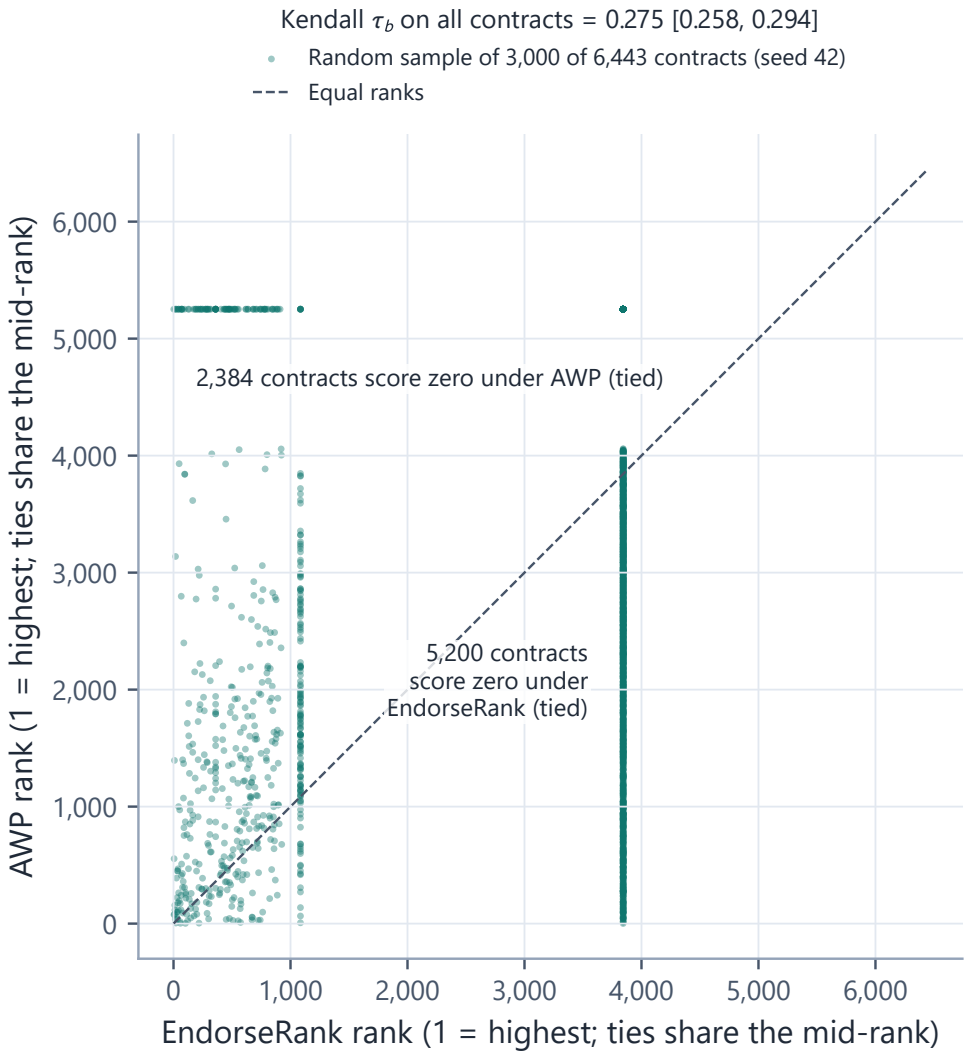


Figure 4.4: Rank–rank scatter of EndorseRank against AWP on the matched cohort (random sample of 3,000 of the 6,443 contracts, seed 42). Tied scores share their average rank, as in Kendall's τ_b. The right-hand column holds the sampled contracts among the 5,200 that score zero under EndorseRank, the column to its left those among the 317 that share the score of a node without in-edges, and the row at the top those among the 2,384 that score zero under AWP. A point on the dashed identity line has the same rank in both orderings; the legend gives τ_b on all contracts with its 95% interval.

Claim C5, is also the premise of the coupled walk (Section 4.8).

4.5 Robustness and sensitivity checks

Three checks ask whether the results of EndorseRank and AWP hinge on one damping value, one slope of the value weight or one sample definition; the third varies the sample itself, so its coefficients are expected to move, and it is not evidence of robustness. Two more test EndorseRank's restart rule and the balance cap.

4.5.1 Damping

Table 4.7 recomputes the family means at $d \in \{0.75, 0.85, 0.95\}$ on the same graphs.

Table 4.7: Robustness: family-mean Kendall τ of EndorseRank (ER) and AWP under PageRank damping $d \in \{0.75, 0.85, 0.95\}$ on the matched cohort ($n = 6{,}443$ contracts); all. = allowance, tr. = transfer, Stab. = stability. Runtimes are means of five timed solver calls on the full graphs; the $d = 0.85$ runtimes are the measurements of Table 4.3, reused, not timed again (Table 8.1 lists every run). CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

d	ER all.	AWP all.	ER tr.	AWP tr.	ER Stab.	AWP Stab.	ER (s)	AWP (s)
0.75	0.839 [0.828, 0.849]	0.224 [0.204, 0.244]	0.297 [0.280, 0.317]	0.750 [0.742, 0.757]	0.220 [0.206, 0.237]	0.643 [0.634, 0.651]	0.086	1.755
0.85	0.839 [0.828, 0.849]	0.227 [0.206, 0.245]	0.297 [0.280, 0.317]	0.755 [0.748, 0.762]	0.220 [0.206, 0.237]	0.643 [0.635, 0.651]	0.101	2.821
0.95	0.839 [0.828, 0.849]	0.228 [0.207, 0.246]	0.297 [0.280, 0.317]	0.761 [0.754, 0.768]	0.220 [0.206, 0.237]	0.641 [0.633, 0.650]	0.208	7.182

For EndorseRank no family mean moves at the third decimal across the three values. For AWP the transfer coefficient rises from 0.750 ($[0.742, 0.757]$) to 0.761 ($[0.754, 0.768]$) as d rises, and its other two family means move by at most 0.004. Neither score's pattern is an artifact of the conventional $d = 0.85$. EndorseRank's stability is largely structural. Its two tie blocks do not depend on d, and for a spender whose owners receive no approvals EndorseRank equals $c\,(1 + d\,\delta(v))$, with $\delta(v)$ the owner-normalized in-strength (Section 5.3.2), so the order among such spenders does not depend on d either. At T_{obs} the formula holds within 1% for all 651 such spenders among the 926 cohort contracts with in-edges, and for 33.8% of the other 275. Runtime grows with the iterations, which

rise with d from 32 to 176 for EndorseRank and from 57 to 294 for AWP, the last just inside the cap of 300; the time ratio is 20.5, 27.9 and 34.5 at an edge ratio of 22.4. Figure 4.5 plots the family means.



Figure 4.5: Damping sensitivity of the family-mean Kendall τ of EndorseRank and AWP on the transfer, allowance and stability families across $d \in \{0.75, 0.85, 0.95\}$, with 95% bootstrap intervals (values from Table 4.7; matched cohort, $n = 6{,}443$ contracts).

4.5.2 Value-slope sensitivity

Table 4.8 recomputes EndorseRank and AWP at T_{obs} with the slope b of the value weight divided and multiplied by ten (Section 3.13.3), on the same graphs and proxies.

Under either slope no family mean moves by more than 0.002, and the pattern stays: EndorseRank leads on the allowance family, and AWP on the transfer and stability families. Each variant orders the contracts almost as the main score of its method does (τ of 0.999 and 0.998 for EndorseRank

Table 4.8: Robustness: family-mean Kendall τ on the matched cohort ($n = 6{,}443$ contracts) when the slope b of the value weight V is divided or multiplied by ten, next to the main analysis. The main slopes put $V = 1/2$ at the median amount, \$160 for transfers and \$0.071 for allowances; $b/10$ moves that point to ten times the median and $10b$ to a tenth of it. The rank agreement τ of each variant with the main score of the same method (EndorseRank and AWP) is 0.999 and 0.986 at $b/10$, and 0.998 and 0.984 at $10b$. No intervals were computed for the variants.

Method	Transfer			Allowance			Stability		
	$b/10$	b	$10b$	$b/10$	b	$10b$	$b/10$	b	$10b$
EndorseRank	0.297	0.297	0.297	0.839	0.839	0.838	0.220	0.220	0.221
AWP	0.754	0.755	0.756	0.227	0.227	0.225	0.642	0.643	0.643

and of 0.986 and 0.984 for AWP).

4.5.3 Sample-definition sensitivity

Table 4.9 recomputes the family means on the subsamples of the runtime scaling and on the full cohort, each scored on the pairs incident to its contracts.

Table 4.9: Sensitivity of the family means to the evaluation set: family-mean Kendall τ of EndorseRank (ER) and AWP on deterministic SHA256 subsamples of the matched cohort (seed `contract-benchmark-v1`, full cohort $n = 6{,}443$ contracts), each row scored on the pairs incident to its contracts, whose numbers the columns $|E_A|$ (allowance) and $|E_T|$ (transfer) give; all. = allowance, tr. = transfer, Stab. = stability. Rows are evaluated on different contract sets and on other graphs than the whole contract graph, so their τ values are not comparable with Table 4.5; they show how much the statistic moves with the evaluation population. No intervals were computed for the rows. Runtimes are the measurements of Table 4.4; the graph of the pairs incident to the full cohort has no timing of its own (–).

n	ER all.	AWP all.	ER tr.	AWP tr.	ER Stab.	AWP Stab.	$\lvert E_A\rvert$	$\lvert E_T\rvert$	ER (s)	AWP (s)
1,000	0.860	0.246	0.322	0.771	0.230	0.666	1,611	120,296	0.001	0.095
2,000	0.853	0.248	0.324	0.762	0.237	0.657	4,788	235,934	0.002	0.134
5,000	0.839	0.235	0.303	0.760	0.226	0.648	11,249	889,794	0.006	0.689
6,443	0.839	0.226	0.297	0.753	0.220	0.643	16,324	1,063,593	–	–

The coefficients are lower at the full cohort than at $n = 1{,}000$: EndorseRank's allowance coefficient from 0.860 to 0.839 and its transfer coefficient from 0.322 to 0.297, AWP's transfer coefficient from 0.771 to 0.753 and its stability coefficient from 0.666 to 0.643. None of these shifts is evidence about the scores. A stage cuts the paths through other contracts, so each score comes closer to a

count of the sampled contract's own edges, which is what the proxies count. The last stage holds the whole cohort with the pairs incident to it, and its family means lie within 0.002 of the main analysis, which scores the cohort on the whole contract graph. The other rows, a sensitivity analysis of the sample definition (Section 3.13.4), must not be compared with Table 4.5.

4.5.4 Restarts of EndorseRank

EndorseRank restarts the walk uniformly, while AWP restarts it at contracts in proportion to their recent sending activity (Section 3.6.2). The analysis plan fixed uniform restarts for EndorseRank and named the variant as a score to report beside it, without fixing any comparison between the two. Table 4.10 scores both versions, with AWP beside them.

Table 4.10: EndorseRank with uniform restarts and with restarts weighted by approving activity, AWP's rule applied to the allowance graph, next to AWP: family-mean Kendall τ on the matched cohort ($n = 6{,}443$ contracts) and τ with the two spender labels in W0 ($n = 7{,}864$ spender contracts, scores frozen at 31 March 2026) and W1 ($n = 7{,}915$, frozen at 30 June 2026). Both EndorseRank rows weight each latest allowance by $\sigma(\Delta t)V(z)$. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

		Same window, matched cohort			W0, spenders		W1, spenders	
Method	Restarts	Transfer	Allowance	Stability	New approval pairs	New senders	New approval pairs	New senders
EndorseRank	uniform	0.297 [0.280, 0.317]	0.839 [0.828, 0.849]	0.220 [0.206, 0.237]	0.168 [0.145, 0.193]	0.084 [0.062, 0.105]	0.147 [0.125, 0.170]	0.064 [0.041, 0.085]
EndorseRank	by approving activity	0.299 [0.281, 0.318]	0.799 [0.782, 0.813]	0.225 [0.210, 0.242]	0.163 [0.142, 0.188]	0.089 [0.069, 0.111]	0.148 [0.126, 0.171]	0.072 [0.049, 0.093]
AWP	by sending activity	0.755 [0.748, 0.762]	0.227 [0.206, 0.245]	0.643 [0.635, 0.651]	0.098 [0.079, 0.119]	0.330 [0.313, 0.347]	0.064 [0.042, 0.084]	0.314 [0.295, 0.332]

With AWP's rule the restart mass goes to the owner contracts in proportion to their approving activity. In the same window EndorseRank's allowance coefficient then falls from 0.839 to 0.799 ($[0.782, 0.813]$), and its transfer coefficient hardly moves (0.299, $[0.281, 0.318]$). Out of window the variant changes little. On new approval pairs it is slightly below the uniform version in W0 (-0.004, $[-0.008, -0.001]$) and level with it in W1 ($+0.001$, $[-0.002, 0.005]$); on new senders it is slightly above it in both windows ($+0.005$, $[0.002, 0.009]$, and $+0.008$, $[0.004, 0.012]$). The W1 contrasts are in Table 4.18; all were computed after the labels were known. Neither version ranks new approval pairs above the in-approve degree (Section 4.7), and Section 4.8.4 sets the variant beside C-PR.

4.5.5 Balance-cap sensitivity

Table 4.11 repeats the out-of-window comparisons on an allowance layer without the cap, in which every positive latest allowance between contracts is an edge at its own value (Section 3.13.5); that layer has $275{,}917$ pairs at t_1 and $302{,}559$ at T_{obs}. It decides no rule.

Table 4.11: Sensitivity to the balance cap: Kendall τ on the two spender labels with the allowance layer capped at the owner's balance (the main analysis) and uncapped, where every positive latest allowance between contracts counts at its own value. Each cell reads capped / uncapped. Without the cap the allowance layer has 275,917 pairs at t_1 against 17,940 with it, and 302,559 against 115,006 at T_{obs}. AWP reads no allowance, so its two values are equal. In the lower block the left cell is the contrast of the main analysis and the right cell the same contrast on the uncapped layer. The uncapped layer decides no rule. Intervals: 95% paired bootstrap (400 resamples, seed 42).

	W0 spenders		W1 spenders	
Score	New approval pairs	New transfer senders	New approval pairs	New transfer senders
EndorseRank	0.168 / 0.088	0.084 / 0.185	0.147 / 0.078	0.064 / 0.177
C-PR ($\lambda = 1$)	0.167 / 0.086	0.084 / 0.183	0.147 / 0.077	0.064 / 0.181
C-PR ($\lambda = 0.5$)	0.142 / 0.150	0.254 / 0.248	0.104 / 0.118	0.236 / 0.234
AWP	0.098 / 0.098	0.330 / 0.330	0.064 / 0.064	0.314 / 0.314
In-approve degree at the freeze	0.204 / 0.105	0.091 / 0.196	0.181 / 0.102	0.072 / 0.189
Union in-degree at the freeze	0.126 / 0.146	0.306 / 0.286	0.093 / 0.118	0.292 / 0.274
Contrasts ($\Delta\tau$ [95% CI]), capped / uncapped				
W0: C-PR minus C-PR ($\lambda = 1$), new approval pairs	-0.025 [-0.045, -0.004]		$+0.064$ [0.051, 0.076]	
W0: C-PR minus C-PR ($\lambda = 0$), new transfer senders	-0.014 [-0.019, -0.008]		-0.019 [-0.027, -0.012]	
W1: C-PR minus C-PR ($\lambda = 0$), new approval pairs	$+0.074$ [0.060, 0.087]		$+0.088$ [0.074, 0.103]	
W1: C-PR minus C-PR ($\lambda = 0$), new transfer senders	-0.016 [-0.022, -0.012]		-0.018 [-0.028, -0.011]	

Without the cap the allowance side has lower coefficients on later approvals and higher ones on later inflow. EndorseRank ranks new approval pairs at 0.088 ($[0.067, 0.107]$) in W0 and 0.078 ($[0.058, 0.099]$) in W1, about half its coefficients with the cap, and new senders at 0.185 ($[0.166, 0.202]$) and 0.177 ($[0.156, 0.197]$), more than twice those with the cap; the in-approve degree moves likewise. No paired contrast between the two layers was computed. C-PR moves little. The contrasts of the rules keep their sign except the contrast with the allowance layer walked alone on new approval pairs, which turns positive in W0 ($+0.064$, $[0.051, 0.076]$, against -0.025 with the cap) and in W1 ($+0.041$, $[0.031, 0.053]$, against -0.043), because the uncapped layer ranks them so much lower. On the uncapped layer C-PR is level with the union in-degree on new approval pairs ($+0.004$, $[-0.004, 0.012]$, in W0 and $+0.000$, $[-0.007, 0.008]$, in W1).

4.6 Discovery order (RQ2)

RQ2 asks whether a crawl ordered by C-PR on the contracts read so far reaches the contracts with the most user-account demand sooner than a breadth-first crawl and than a crawl ordered by the number of contract backlinks. The discovery-order test (Section 3.7) repeats the crawl-ordering experiment of Cho et al. (1998) on the contract graph. It was defined after the registered results were known and decides no rule of the analysis plan or the registration; its own decision rule is read in one run, at T_{obs} in the primary reveal mode. In that mode, `both`, a visit reveals a contract's in-edges and out-edges, and the universe $\mathcal{U}$ is the weakly connected component that holds the giant strongly connected component, $1{,}169{,}044$ contracts. The protocol's rule made it primary because the other universe, the giant strongly connected component with the contracts reachable from it (SCC$\cup$OUT), holds fewer than half as many contracts in the structure pilot at t_1 (Table 8.7 in Appendix 8.4.4). The pilot also raised the protocol's stop flag on a thin allowance layer (Section 3.7); what the allowance layer adds is read from the contrasts outside the decision. Importance is the demand of user accounts, which no ordering reads: the dollars they sent to a contract in the window (I_T) and the balance-capped dollars they had authorized it to spend at the freeze (I_A). The universe holds 96.5% of I_T and 93.0% of I_A (Table 8.8) and $16{,}776$ of the capped allowance pairs, so the pairs of the largest owner lie outside it. Its hot sets hold $13{,}141$ contracts by I_T and $11{,}662$ by I_A; each ordering crawls from 20 seeds, and $\text{AUC}_{10\%}$ measures how early it visits the hot contracts.

Table 4.12 gives the two decision contrasts. D_1, C-PR′ minus the breadth-first crawl, is $+0.0520$ ($[+0.0485, +0.0558]$) on I_T and $+0.0061$ ($[+0.0014, +0.0111]$) on I_A, both intervals above zero. D_2, C-PR′ minus the crawl ordered by contract backlinks, is -0.0051 ($[-0.0087, -0.0010]$) on I_T and -0.0571 ($[-0.0621, -0.0519]$) on I_A, both intervals below zero. The verdict is V2 on both importances, with the sentence that an interval below zero adds, and the statement of the test reads: "Ordered by C-PR on the contracts read so far, a crawl from a random contract of the giant strongly connected component reached the contracts of the universe with the most user-account inflow and those with the most user-authorised value sooner than a breadth-first crawl, but not sooner than a

crawl ordered by the number of contract backlinks. A crawl ordered by the number of contract backlinks reached them sooner." The answer to RQ2 is therefore no, and the lead over the breadth-first crawl is small on I_A.

Table 4.12: Discovery-order test: the two decision contrasts on each importance, $D_1 = \mathrm{AUC}_{10\%}(\text{C-PR}') - \mathrm{AUC}_{10\%}(\text{BFS})$ and $D_2 = \mathrm{AUC}_{10\%}(\text{C-PR}') - \mathrm{AUC}_{10\%}(\mathrm{IB}'_U)$, with 95% percentile intervals of a paired bootstrap over the hot contracts (400 resamples, seed 42) and the shares of resamples above 0 for D_1 and D_2. The decision is read at T_{obs} in the primary mode only. Its verdict on an importance is V1 when both intervals lie above 0, V2 when the interval of D_1 lies above 0 and that of D_2 does not, and V3 when the interval of D_1 does not lie above 0; an interval of D_2 below 0 adds that the crawl ordered by contract backlinks was sooner. The answer to the question of the test is the weaker of the two verdicts, so it is yes only when both are V1. The other three runs carry no verdict: for each importance the contrasts whose interval lies above 0 and those whose interval lies below 0 are listed. The two freezes share nine months of their score windows and are not independent. The test was defined after the registered results were known; it decides no rule of the analysis plan or the registration.

Importance	$\lvert H_q\rvert$	D_1	95% interval	D_2	95% interval	Share above 0 (D_1 / D_2)	Reading
T_{obs}, `both` mode: the decision							
I_T	13,141	+0.0520	[+0.0485, +0.0558]	-0.0051	[-0.0087, -0.0010]	1.000 / 0.005	V2: sooner than BFS, not sooner than IB'_U; IB'_U sooner
I_A	11,662	+0.0061	[+0.0014, +0.0111]	-0.0571	[-0.0621, -0.0519]	0.998 / 0.000	V2: sooner than BFS, not sooner than IB'_U; IB'_U sooner
T_{obs}, `out` mode: no verdict							
I_T	5,087	+0.2650	[+0.2540, +0.2743]	+0.0803	[+0.0690, +0.0920]	1.000 / 1.000	above 0: D_1, D_2; below 0: none
I_A	5,088	+0.1143	[+0.1053, +0.1237]	+0.0436	[+0.0347, +0.0522]	1.000 / 1.000	above 0: D_1, D_2; below 0: none
t_1, `both` mode: no verdict							
I_T	12,697	+0.0574	[+0.0530, +0.0611]	-0.0158	[-0.0196, -0.0121]	1.000 / 0.000	above 0: D_1; below 0: D_2
I_A	8,136	+0.0230	[+0.0164, +0.0291]	-0.1010	[-0.1083, -0.0933]	1.000 / 0.000	above 0: D_1; below 0: D_2
t_1, `out` mode: no verdict							
I_T	3,324	+0.3414	[+0.3282, +0.3547]	+0.1222	[+0.1079, +0.1361]	1.000 / 1.000	above 0: D_1, D_2; below 0: none
I_A	3,324	+0.1705	[+0.1564, +0.1822]	+0.0739	[+0.0607, +0.0848]	1.000 / 1.000	above 0: D_1, D_2; below 0: none

Table 4.13 gives the level of every ordering and Figure 4.6 the curves. In the deciding run the breadth-first crawl has an $\mathrm{AUC}_{10\%}$ of 0.0565 on I_T and 0.0882 on I_A, C-PR′ 0.1085 and 0.0943, and IB'_U 0.1136 and 0.1514. AWP′ has the highest value of the crawls on both, 0.1465 and 0.1582, apart from the diagnostic δ'. Every crawl stays far from the ideal order, which reaches 0.9438 and 0.9501 (the random curve 0.0500). The partial view is not the only cause: ordered with the full graph in view, C-PR reaches 0.1061 and 0.1157 and the union in-degree 0.1593 and 0.1994.

Table 4.14 gives the contrasts outside the decision in the order the protocol fixes. First, each walk is set against the count it smooths. AWP′ is the one single-layer walk above its count on both importances ($+0.0329$, $[+0.0298, +0.0368]$, and $+0.0116$, $[+0.0071, +0.0162]$, over IB'_T). The walks on the allowance layer alone lie at or below IB'_A, except EndorseRank-activity′ on I_A; C-PR′

Table 4.13: Discovery-order test: $\mathrm{AUC}_{10\%}$ of every ordering on the hot sets of the two importances, user-account inflow I_T and user-authorized value I_A, in the four runs. $\mathrm{AUC}_{10\%}$ is the mean over the hot contracts of $\max(0, 1 - t_h/\lceil 0.1|\mathcal{U}|\rceil)$, where $\mathcal{U}$ is the universe of the run and t_h the visit at which the crawl reaches hot contract h, averaged over 20 seeds from the giant strongly connected component; a crawl recomputes its ordering each time its visits have grown by the share $\rho = 1/50$. The first two columns (T_{obs}, `both` mode) are the run that the decision reads (Table 4.12). A primed name is the count or walk computed on the contracts read so far. δ' is a diagnostic outside every comparison. The references are not crawls: the orders by C-PR and by union in-degree of the full graph, the ideal order (hot contracts first) and the random curve. The test was defined after the registered results were known; it decides no rule of the analysis plan or the registration.

	T_{obs}, `both`		T_{obs}, `out`		t_1, `both`		t_1, `out`	
Ordering	I_T	I_A	I_T	I_A	I_T	I_A	I_T	I_A
Breadth-first (BFS)	0.0565	0.0882	0.1179	0.0595	0.0543	0.1088	0.1269	0.0595
Random	0.0609	0.0725	0.1085	0.0621	0.0543	0.0930	0.1169	0.0703
IB'_A (allowance backlinks)	0.0626	0.0881	0.1438	0.0818	0.0539	0.1094	0.1750	0.1076
IB'_T (transfer backlinks)	0.1135	0.1466	0.3023	0.1233	0.1107	0.2001	0.3460	0.1478
IB'_U (contract backlinks)	0.1136	0.1514	0.3026	0.1302	0.1276	0.2328	0.3461	0.1561
EndorseRank′	0.0621	0.0869	0.1404	0.0802	0.0526	0.1062	0.1733	0.1068
EndorseRank-activity′	0.0612	0.1042	0.1404	0.0802	0.0587	0.1426	0.1734	0.1066
AWP′	0.1465	0.1582	0.4446	0.1753	0.1524	0.2220	0.4766	0.2231
C-PR′ ($\lambda = 1$)	0.0617	0.0864	0.1398	0.0789	0.0526	0.1057	0.1728	0.1063
C-PR′ ($\lambda = 0$)	0.1061	0.0806	0.4133	0.1584	0.1088	0.1097	0.4818	0.2020
C-PR′ ($\lambda = 0.25$)	0.1081	0.0930	0.4129	0.1835	0.1110	0.1297	0.4817	0.2349
C-PR′ ($\lambda = 0.5$)	0.1085	0.0943	0.3829	0.1738	0.1118	0.1319	0.4683	0.2300
C-PR′ ($\lambda = 0.75$)	0.1091	0.0952	0.3744	0.1608	0.1121	0.1323	0.4562	0.2197
S-PR′	0.1069	0.0861	0.4190	0.1805	0.1095	0.1170	0.4874	0.2360
δ' (diagnostic, outside the comparisons)	0.1643	0.1942	0.5357	0.2317	0.1775	0.2778	0.5795	0.2937
Reference: C-PR of the full graph	0.1061	0.1157	0.2449	0.1279	0.1165	0.1646	0.3239	0.1773
Reference: union in-degree of the full graph	0.1593	0.1994	0.4600	0.2142	0.1757	0.3151	0.5826	0.2846
Reference: ideal order	0.9438	0.9501	0.9500	0.9500	0.9375	0.9600	0.9500	0.9500
Reference: random curve	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500

($\lambda = 0$) lies below IB'_T; and C-PR′ lies above IB'_A and below IB'_T. Second, among the walks, the allowance layer adds a small, resolved amount to C-PR′ over its transfer layer walked alone ($+0.0024$, $[+0.0016, +0.0033]$, and $+0.0137$, $[+0.0117, +0.0156]$), and C-PR′ lies below AWP′ (-0.0379, $[-0.0412, -0.0347]$, and -0.0639, $[-0.0683, -0.0591]$). Third, $\lambda = 0.25$ and 0.75 give the same reading. Fourth, on hot sets defined on the edges that the orderings read, each ordering finds its own target, an intended-construct check: on the hot set of the union in-degree the backlink

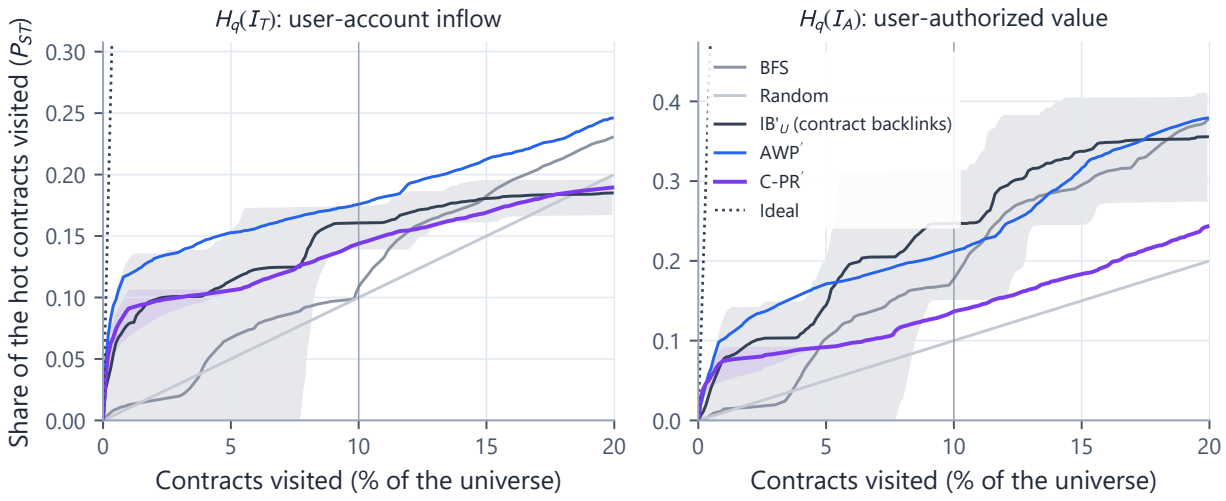


Figure 4.6: Discovery-order test at T_{obs} in the primary mode: the share of the hot contracts visited, P_{ST}, against the contracts visited as a percentage of the universe, for the hot sets of user-account inflow I_T (left) and of user-authorized value I_A (right). The curves are means over the 20 seeds for the breadth-first and random orderings, the backlink count IB'_U, AWP′ and C-PR′; the shaded bands give the minimum and maximum over the seeds for IB'_U and C-PR′, the dotted line is the ideal order, and the vertical line marks 10% of the universe, the range of $\text{AUC}_{10\%}$. The test was defined after the registered results were known; it decides no rule of the analysis plan or the registration.

crawl is far sooner than C-PR′ (-0.3031, $[-0.3106, -0.2958]$), and on the hot set of full-graph C-PR, C-PR′ is far sooner ($+0.2606$, $[+0.2544, +0.2663]$). The first of these repeats the comparison of Cho et al. (1998), in which the PageRank ordering was sooner; here it goes the other way.

Fifth, the sensitivities change single cells and not the reading that C-PR′ does not beat the backlink count (Table 8.9). No row puts the interval of D_2 above zero. D_1 on I_A, the smallest decision contrast, does not hold throughout: it lies below zero with the area taken over 20% of the universe and under a schedule that recomputes four times as often, and its interval contains zero without the hubs and under a bootstrap over the seeds. The other reveal mode belongs to these sensitivities and carries no verdict. In the `out` mode, where a visit reveals out-edges only and the universe is SCC ∪ OUT, all four contrasts lie above zero at both freezes (Table 4.12): at T_{obs} D_2 is $+0.0803$ ($[+0.0690, +0.0920]$) on I_T and $+0.0436$ ($[+0.0347, +0.0522]$) on I_A, and at t_1 it is larger. The result therefore depends on what a visit reveals (Figure 8.1). Sixth, at t_1 in the primary mode the intervals of D_1 lie above zero and those of D_2 below zero on both importances, as at T_{obs}; the two

Table 4.14: Discovery-order test at T_{obs} in the primary mode: the contrasts reported outside the decision, in the order fixed in the protocol. Each cell is the difference of $\text{AUC}_{10\%}$ between two orderings with its 95% percentile intervals of a paired bootstrap over the hot contracts (400 resamples, seed 42); an interval above 0 means the first ordering reached the hot contracts sooner. Block 1 sets each walk beside the count it smooths; block 2 compares the walks, first C-PR′ with its transfer layer walked alone, C-PR′ ($\lambda = 0$), and with AWP′, which shows what the allowance layer adds. In block 4 the hot sets are defined on the edges that the orderings read, the union in-degree IB_U and C-PR of the full graph (IR), so agreement there is an intended-construct check; C-PR′ minus IB'_U on $H_q(\text{IB}_U)$ is the contrast that repeats the comparison of Cho et al. The test was defined after the registered results were known; it decides no rule of the analysis plan or the registration.

Contrast	On $H_q(I_T)$	On $H_q(I_A)$
1. Each walk′ against the count it smooths		
EndorseRank′ − IB'_A	-0.0006 [-0.0007, -0.0004]	-0.0012 [-0.0014, -0.0010]
EndorseRank-activity′ − IB'_A	-0.0014 [-0.0030, +0.0001]	+0.0161 [+0.0136, +0.0185]
C-PR′ ($\lambda = 1$) − IB'_A	-0.0009 [-0.0010, -0.0007]	-0.0017 [-0.0019, -0.0015]
AWP′ − IB'_T	+0.0329 [+0.0298, +0.0368]	+0.0116 [+0.0071, +0.0162]
C-PR′ ($\lambda = 0$) − IB'_T	-0.0074 [-0.0109, -0.0031]	-0.0660 [-0.0709, -0.0609]
C-PR′ − IB'_A	+0.0459 [+0.0427, +0.0492]	+0.0062 [+0.0023, +0.0106]
C-PR′ − IB'_T	-0.0050 [-0.0086, -0.0008]	-0.0523 [-0.0575, -0.0468]
2. Among the walks		
C-PR′ − C-PR′ ($\lambda = 0$)	+0.0024 [+0.0016, +0.0033]	+0.0137 [+0.0117, +0.0156]
C-PR′ − AWP′	-0.0379 [-0.0412, -0.0347]	-0.0639 [-0.0683, -0.0591]
C-PR′ − C-PR′ ($\lambda = 1$)	+0.0468 [+0.0436, +0.0501]	+0.0078 [+0.0040, +0.0123]
C-PR′ − EndorseRank′	+0.0465 [+0.0433, +0.0498]	+0.0074 [+0.0035, +0.0117]
S-PR′ − C-PR′ ($\lambda = 0$)	+0.0008 [+0.0006, +0.0011]	+0.0054 [+0.0044, +0.0066]
3. Other values of λ		
C-PR′ ($\lambda = 0.25$) − BFS	+0.0515 [+0.0480, +0.0555]	+0.0049 [+0.0002, +0.0100]
C-PR′ ($\lambda = 0.25$) − IB'_U	-0.0055 [-0.0091, -0.0016]	-0.0584 [-0.0633, -0.0531]
C-PR′ ($\lambda = 0.25$) − C-PR′ ($\lambda = 0.5$)	-0.0005 [-0.0008, -0.0001]	-0.0012 [-0.0017, -0.0008]
C-PR′ ($\lambda = 0.75$) − BFS	+0.0526 [+0.0491, +0.0563]	+0.0070 [+0.0024, +0.0121]
C-PR′ ($\lambda = 0.75$) − IB'_U	-0.0045 [-0.0080, -0.0005]	-0.0562 [-0.0610, -0.0508]
C-PR′ ($\lambda = 0.75$) − C-PR′ ($\lambda = 0.5$)	+0.0006 [+0.0002, +0.0010]	+0.0009 [+0.0005, +0.0014]
4. Hot sets in the manner of Cho et al. (an intended-construct check)		
C-PR′ − BFS on $H_q(\text{IB}_U)$	+0.0176 [+0.0119, +0.0238]	
C-PR′ − IB'_U on $H_q(\text{IB}_U)$	-0.3031 [-0.3106, -0.2958]	
C-PR′ − BFS on $H_q(\text{IR})$	+0.3084 [+0.3024, +0.3143]	
C-PR′ − IB'_U on $H_q(\text{IR})$	+0.2606 [+0.2544, +0.2663]	

freezes share nine months of their score windows and are not independent.

The diagnostics show how the crawls differ (Table 8.10). The count orderings decide nearly every pick by a tie (0.9998 to 0.9999 of the picks), as do the walks on the allowance layer alone, and the order of discovery breaks a tie, as in a breadth-first crawl. C-PR′ decides 0.535 of its picks by a tie in the primary mode and 0.023 in the `out` mode, AWP′ 0.252 and 0.005. Under every walk and count, 0.84 to 0.91 of the first tenth of the visits go to contracts with no user-account demand.

4.7 Temporal holdout of the single-layer walks (RQ3)

The same-window check cannot separate a score from its own construct, and the discovery-order test reads importance in the score window; the out-of-window evaluation tests whether a score frozen at a cut-off predicts what happens after it. Following Section 3.8, scores on the W0 spender cohort are frozen at t_1, and two labels are counted for each contract between 1 April and 30 June 2026: new owner–spender approval pairs and new transfer senders. Table 4.15 sets every score against both labels, with four raw counts as baselines: the in-approve degree and value, the transfer in-degree and the union in-degree. Table 4.16 gives the paired contrasts: the two of rule A, fixed in the analysis plan, and those computed after the labels were known. This section reads each walk against the count it smooths; rule A and the other rows of C-PR and S-PR are taken up in Section 4.8.

Two facts frame every out-of-window number. The labels count every account: 90.4% of the $748{,}258$ new approval pairs of W0 come from user accounts, and 86.3% of its $651{,}474$ new senders are user accounts (in W1 87.9% and 82.3%; Table 8.6 in Appendix 8.4.3), and no score reads an edge of a user account. And 27.5% of the W0 spenders and 27.2% of the W1 spenders lie outside the union graph and score zero under every score and count (Table 4.2). Both labels are also sparse: of the 7,864 spenders of W0, 1,725 gained a new approval pair and 1,416 a new sender. Each label is one kind of event carried into the future, so these results show whether a score predicts more of the activity it summarizes and not whether a contract is safe.

Table 4.15: Temporal holdout W0 on the spender cohort ($n = 7{,}864$ contracts): Kendall τ between scores frozen at 31 March 2026 on the contract graph and two labels counted on the contract between 1 April 2026 and 30 June 2026: the number of new owner–spender approval pairs and the number of accounts that sent a transfer to the contract for the first time. The cohort keeps the 7,864 contracts among the 7,962 spenders of the labels; 97 user accounts and 1 system address are left out. 1,725 contracts received at least one new pair and 1,416 at least one new sender. 5,700 of the cohort (72.5%) lie in the union graph; a contract outside a graph scores zero. Baseline rows are raw counts on the contract graph at t_1: the in-approve degree and value and the transfer in-degree that the single-layer walks smooth, and the union in-degree, the number of distinct contracts with an edge to the contract in either layer. C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$) are the allowance and transfer layers of the coupled operator walked alone. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Score at t_1	New approval pairs (τ [95% CI])	New transfer senders (τ [95% CI])
In-approve degree at t_1 (baseline)	0.204 [0.181, 0.228]	0.091 [0.066, 0.113]
In-approve value at t_1 (baseline)	0.189 [0.166, 0.213]	0.081 [0.057, 0.103]
Transfer in-degree at t_1 (baseline)	0.070 [0.050, 0.091]	0.318 [0.298, 0.336]
Union in-degree at t_1 (baseline)	0.126 [0.108, 0.146]	0.306 [0.287, 0.324]
EndorseRank	0.168 [0.145, 0.193]	0.084 [0.062, 0.105]
AWP	0.098 [0.079, 0.119]	0.330 [0.313, 0.347]
C-PR ($\lambda = 1$)	0.167 [0.145, 0.192]	0.084 [0.062, 0.105]
C-PR ($\lambda = 0$)	0.070 [0.052, 0.091]	0.268 [0.250, 0.285]
C-PR ($\lambda = 0.25$)	0.141 [0.123, 0.161]	0.256 [0.238, 0.273]
C-PR ($\lambda = 0.5$)	0.142 [0.124, 0.162]	0.254 [0.235, 0.270]
C-PR ($\lambda = 0.75$)	0.142 [0.124, 0.162]	0.251 [0.232, 0.267]
S-PR	0.091 [0.074, 0.113]	0.264 [0.246, 0.283]

The first finding answers the second part of RQ3 for the single-layer walks, and it differs by layer. On the allowance side the count wins. The in-approve degree at t_1 ranks new approval pairs at $\tau = 0.204$ ($[0.181, 0.228]$) and EndorseRank at 0.168 ($[0.145, 0.193]$), a paired difference of -0.036 ($[-0.046, -0.027]$); C-PR ($\lambda = 1$) lies as far below the count (-0.037, $[-0.046, -0.028]$), and so does EndorseRank with activity restarts (-0.041, $[-0.053, -0.029]$). On the transfer side AWP ranks new transfer senders at 0.330 ($[0.313, 0.347]$), slightly above the transfer in-degree at 0.318 ($[0.298, 0.336]$): the paired difference is $+0.012$ ($[0.003, 0.023]$). The transfer layer walked with uniform restarts lies lower, C-PR ($\lambda = 0$) at 0.268 ($[0.250, 0.285]$). These degree contrasts were computed after the labels were known. The coupled walk smooths the union in-degree, the count of the edges of both layers, and against it the result is split: C-PR ranks new approval pairs higher

Table 4.16: Paired bootstrap differences on the spender holdout W0 ($n = 7{,}864$ contracts; 400 resamples shared by a and b, seed 42). The first block holds the two contrasts of the primary part of rule A, fixed in the analysis plan before the study extracted any data and applied here to the scores of the contract graph (C-PR against C-PR ($\lambda = 1$) on new approval pairs and against C-PR ($\lambda = 0$) on new transfer senders). The second block was computed after the labels were known: the increment of C-PR ($\lambda = 1$), the allowance layer walked alone, over its raw degree at t_1, the same comparisons for S-PR, and the increments of EndorseRank and AWP over their raw degrees; C-PR's other contrasts are in Table 4.22. The third block sets C-PR beside the union in-degree. Primary part of rule A (C-PR no worse than C-PR ($\lambda = 1$) on new approval pairs and better than C-PR ($\lambda = 0$) on new transfer senders): no worse fails, better fails, so the rule is not met. The differences between EndorseRank and AWP are in Table 8.15.

Contrast (a minus b)	Label	$\tau(a)$	$\tau(b)$	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
Rule A, primary part, fixed in the plan						
C-PR minus C-PR ($\lambda = 1$)	new approval pairs	0.142	0.167	-0.025	[-0.045, -0.004]	0.015
C-PR minus C-PR ($\lambda = 0$)	new transfer senders	0.254	0.268	-0.014	[-0.019, -0.008]	0.003
Computed after the labels were known						
C-PR ($\lambda = 1$) minus t_1 in-approve degree	new approval pairs	0.167	0.204	-0.037	[-0.046, -0.028]	0.000
S-PR minus C-PR ($\lambda = 1$)	new approval pairs	0.091	0.167	-0.077	[-0.098, -0.055]	0.000
S-PR minus C-PR ($\lambda = 0$)	new transfer senders	0.264	0.268	-0.003	[-0.013, 0.008]	0.285
EndorseRank minus t_1 in-approve degree	new approval pairs	0.168	0.204	-0.036	[-0.046, -0.027]	0.000
AWP minus t_1 in-degree	new transfer senders	0.330	0.318	+0.012	[0.003, 0.023]	0.998
C-PR against the union in-degree, the baseline of the contract graph						
C-PR minus t_1 union in-degree	new approval pairs	0.142	0.126	+0.016	[0.007, 0.024]	1.000
C-PR minus t_1 union in-degree	new transfer senders	0.254	0.306	-0.053	[-0.061, -0.043]	0.000

($+0.016$, $[0.007, 0.024]$) and new senders lower (-0.053, $[-0.061, -0.043]$).

AWP's lead holds among the W0 spenders inside the union graph ($+0.021$, $[0.005, 0.038]$) but not among the contracts that the count can rank by their inflow. Table 4.17 restricts the holdout to the 4,569 spenders that had received a transfer from another contract before the freeze. There AWP ranks new senders below the transfer in-degree (0.411, $[0.389, 0.429]$, against 0.446, $[0.423, 0.467]$; -0.035, $[-0.049, -0.022]$), and EndorseRank stays below the in-approve degree on new approval pairs (-0.045, $[-0.058, -0.033]$). The two results for AWP are compatible: the transfer in-degree is zero for the other 3,295 spenders, whereas AWP, which restarts at senders, is zero only for the 2,421 without a transfer pair and sets the remaining 874 above them. On new approval pairs AWP reaches 0.238 ($[0.219, 0.260]$) on the subset, above the point estimates of EndorseRank (0.167, $[0.139, 0.197]$) and of the in-approve degree (0.212, $[0.183, 0.242]$), so the lead of the allowance side on this label rests on the spenders outside the subset. The restriction was chosen after the

results were known. No paired contrast was computed for the three estimates on the subset, for the 3,295 alone, or for C-PR ($\lambda = 0$) against its count.

Table 4.17: Spender holdout W0 restricted to the 4,569 of the 7,864 spender contracts that had received a transfer from another contract before the freeze ($n = 4{,}569$). The other 3,295 have no inbound transfer edge from another contract, and 2,421 spenders score zero under AWP. The restriction was chosen after the results were known. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42). [a]Computed afterwards from the saved scores, with the same resamples; its contrasts with AWP and the in-approve degree on this subset are in Table 4.22.

Score at t_1	New approval pairs (τ [95% CI])	New transfer senders (τ [95% CI])
EndorseRank	0.167 [0.139, 0.197]	0.068 [0.040, 0.094]
AWP	0.238 [0.219, 0.260]	0.411 [0.389, 0.429]
C-PR ($\lambda = 0.5$)[a]	0.264 [0.244, 0.286]	0.327 [0.306, 0.347]
In-approve degree at t_1	0.212 [0.183, 0.242]	0.104 [0.076, 0.133]
Transfer in-degree at t_1	0.220 [0.196, 0.243]	0.446 [0.423, 0.467]
EndorseRank minus t_1 in-approve degree (new approval pairs)	$\Delta\tau = -0.045$ [-0.058, -0.033]	
AWP minus t_1 in-degree (new transfer senders)	$\Delta\tau = -0.035$ [-0.049, -0.022]	

Read across the whole spender cohort, each label is ranked best by the scores and counts of its own edge type. On new approval pairs the in-approve degree leads, followed by the in-approve value, EndorseRank and C-PR ($\lambda = 1$), then the C-PR mixtures and the union in-degree, and the transfer side stays lowest (AWP 0.098, $[0.079, 0.119]$). On new transfer senders AWP, the transfer in-degree and the union in-degree lead, and the allowance side stays below 0.10 (EndorseRank 0.084, $[0.062, 0.105]$). The reach of the layers is part of this. EndorseRank scores $5{,}966$ of the $7{,}864$ spenders zero and AWP $2{,}421$; the spenders without a transfer pair gained new approval pairs about as often as the rest (23.4% against 21.3%) and new senders far less often (7.9% against 22.5%), so AWP's zeros cost it little on the first label and help it on the second. Within a three-month window a spender gains more new owners and senders the more it already has, as preferential attachment would predict (Barabási & Albert, 1999), and a local count captures this directly.

Two other candidate labels are reported without being offered as support. Revocation counts, which for some tokens include allowances spent in full (Section 3.8), and the drain heuristic both rise with every score, most steeply with the raw in-approve degree ($\tau = 0.190$, $[0.161, 0.220]$, and 0.213, $[0.188, 0.238]$). For an endorsement reading the sign is wrong, since a trusted spender should lose

approvals less often; a spender with many owners sees more of everything owners do, so these labels cannot separate trust from use.

4.7.1 The single-layer walks in the registered window W1

The registered window W1 (Section 3.8.1) freezes the scores at T_{obs}, 30 June 2026, and counts the same two labels from July to September 2026; rule B, which concerns C-PR, is reported in Section 4.8.3. Of the 7,915 spender contracts, 1,491 gained a new approval pair in the label window and 1,369 a new sender. Table 4.18 gives EndorseRank and its variant with activity restarts, which the registration does not name, beside AWP, C-PR and the two degrees. None of its rows enters a rule, and its contrasts were computed after the labels were known.

The W0 pattern recurs in this window, on an allowance layer of another shape (Section 4.1). The in-approve degree at the freeze again ranks new approval pairs above EndorseRank (0.181, $[0.158, 0.208]$, against 0.147, $[0.125, 0.170]$; -0.034, $[-0.043, -0.025]$), and with activity restarts EndorseRank lies as far below it (-0.033, $[-0.044, -0.022]$). AWP again ranks new senders slightly above the transfer in-degree (0.314, $[0.295, 0.332]$, against 0.302, $[0.278, 0.323]$; $+0.011$, $[0.001, 0.022]$); the restriction to the spenders that had received a transfer from another contract was not computed for this contrast in W1. C-PR again lies above the union in-degree on new approval pairs ($+0.011$, $[0.003, 0.019]$) and below it on new senders (-0.056, $[-0.065, -0.047]$; Table 4.22).

4.8 The coupled walk C-PR (RQ4, RQ5)

C-PR, the method of this thesis, walks both US-dollar layers and mixes their transition rows at each node with weight λ (Section 3.6.3). The analysis plan fixed one decision rule, and its object is C-PR: rule A asks the coupled walk to beat each of its layers walked alone (RQ4). The registration of W1 added rule B, which asks it to improve on its transfer layer on new data (RQ5). Both rules compare C-PR with its own layers, C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$), because only that contrast changes the coupling and nothing else; C-PR also differs from EndorseRank and AWP in its amount weights

Table 4.18: EndorseRank and its variant with activity restarts on the spender cohort and labels of window W1 ($n = 7{,}915$ contracts). Neither these rows nor the contrasts below them belong to the two applied criteria of rule B, and the contrasts were not named in the registration. The contrasts of EndorseRank with activity restarts were computed afterwards from the saved W1 scores. The AWP, C-PR and degree rows repeat Table 4.20, and every contrast uses the same contract resamples; the differences between EndorseRank and AWP are in Table 8.15, and those of C-PR in Table 4.22. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Score at 30 June 2026	New approval pairs (τ [95% CI])	New transfer senders (τ [95% CI])
In-approve degree at the freeze (baseline)	0.181 [0.158, 0.208]	0.072 [0.046, 0.096]
Transfer in-degree at the freeze (baseline)	0.031 [0.008, 0.052]	0.302 [0.278, 0.323]
EndorseRank	0.147 [0.125, 0.170]	0.064 [0.041, 0.085]
EndorseRank, restarts by approving activity	0.148 [0.126, 0.171]	0.072 [0.049, 0.093]
AWP	0.064 [0.042, 0.084]	0.314 [0.295, 0.332]
C-PR ($\lambda = 0.5$)	0.104 [0.083, 0.126]	0.236 [0.217, 0.255]
EndorseRank minus in-approve degree, new approval pairs	$\Delta\tau = -0.034$ [-0.043, -0.025]	
AWP minus transfer in-degree, new transfer senders	$\Delta\tau = +0.011$ [0.001, 0.022]	
EndorseRank (activity restarts) minus in-approve degree, new approval pairs	$\Delta\tau = -0.033$ [-0.044, -0.022]	
EndorseRank (activity restarts) minus EndorseRank, new approval pairs	$\Delta\tau = +0.001$ [-0.002, 0.005]	
EndorseRank (activity restarts) minus EndorseRank, new transfer senders	$\Delta\tau = +0.008$ [0.004, 0.012]	

and its uniform restarts. Both are applied to the scores of the contract graph as a re-analysis. The contrasts of C-PR with every other score and count, at the end of the section, decide no rule.

4.8.1 The coupled walk within the window

Table 4.19 gives the paired contrasts in this window of C-PR and its ablation S-PR, which keeps the transfer layer but restarts from the allowance walk (Section 3.6.3): first the eight of Section 3.11 against the two layers walked alone, then six against EndorseRank and AWP, two of them computed afterwards from the saved scores. RQ4 and RQ5 are decided out of window; this window only locates the hybrids.

C-PR lies between its two layers on every family and stays nearer to its transfer layer. At the primary setting it reaches 0.738 ($[0.729, 0.747]$) on the transfer family, below C-PR ($\lambda = 0$) by $\Delta\tau = -0.066$ ($[-0.074, -0.058]$), and 0.433 ($[0.419, 0.445]$) on the allowance family, below C-PR ($\lambda = 1$) by $\Delta\tau = -0.406$ ($[-0.418, -0.393]$). The gap on the allowance family is in part one of reach. C-PR gives a positive score to the $4{,}270$ cohort contracts in the union graph, more than four times the 926 that the allowance layer reaches as spenders, and so orders by their transfers

Table 4.19: Same-window paired contrasts for the coupled operator (C-PR, $\lambda = 0.5$) and the endorsement-seeded walk (S-PR) against the two layers they are built from, the allowance layer (C-PR ($\lambda = 1$)) and the transfer layer (C-PR ($\lambda = 0$)), and of C-PR against EndorseRank and AWP (matched cohort, $n = 6{,}443$ contracts). Family-mean τ; the same contract resamples are used for a and b. The contrasts with the layers are a stricter check beside the secondary part of rule A, which compares C-PR's point estimate with the better layer's interval in Table 4.5. The secondary part of rule A, fixed in the analysis plan and applied here to the scores of the contract graph, is not met: C-PR's point estimate lies outside the interval of C-PR ($\lambda = 0$) on the transfer family and outside the interval of C-PR ($\lambda = 1$) on the allowance family, and C-PR does not exceed both layers on the stability family. An interval below 0 means a is below b beyond sampling variation; one above 0 means it exceeds it. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42). [a]Computed afterwards from the saved scores, with the same resamples: C-PR against EndorseRank on the transfer family and against AWP on the allowance family, the family that each single-layer walk follows least.

Contrast (a minus b)	$\tau(a)$	$\tau(b)$	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
Against the two layers walked alone					
C-PR transfer minus C-PR ($\lambda = 0$) transfer	0.738	0.804	-0.066	[-0.074, -0.058]	0.000
C-PR allowance minus C-PR ($\lambda = 1$) allowance	0.433	0.838	-0.406	[-0.418, -0.393]	0.000
C-PR stability minus C-PR ($\lambda = 0$) stability	0.574	0.637	-0.063	[-0.071, -0.053]	0.000
C-PR stability minus C-PR ($\lambda = 1$) stability	0.574	0.222	+0.352	[0.340, 0.365]	1.000
S-PR transfer minus C-PR ($\lambda = 0$) transfer	0.782	0.804	-0.022	[-0.030, -0.013]	0.000
S-PR allowance minus C-PR ($\lambda = 1$) allowance	0.371	0.838	-0.467	[-0.488, -0.446]	0.000
S-PR stability minus C-PR ($\lambda = 0$) stability	0.584	0.637	-0.053	[-0.058, -0.048]	0.000
S-PR stability minus C-PR ($\lambda = 1$) stability	0.584	0.222	+0.362	[0.346, 0.379]	1.000
Against EndorseRank and AWP					
C-PR transfer minus AWP transfer	0.738	0.755	-0.017	[-0.028, -0.006]	0.000
C-PR allowance minus EndorseRank allowance	0.433	0.839	-0.406	[-0.419, -0.393]	0.000
C-PR stability minus AWP stability	0.574	0.643	-0.069	[-0.077, -0.059]	0.000
C-PR stability minus EndorseRank stability	0.574	0.220	+0.353	[0.341, 0.367]	1.000
C-PR transfer minus EndorseRank transfer[a]	0.738	0.297	+0.441	[0.423, 0.457]	1.000
C-PR allowance minus AWP allowance[a]	0.433	0.227	+0.206	[0.188, 0.224]	1.000

more than three thousand contracts that the allowance proxies tie at zero, which τ_b counts against it. AWP ranks the transfer family better than C-PR (-0.017, $[-0.028, -0.006]$), and EndorseRank the allowance family (-0.406, $[-0.419, -0.393]$). On the family that each single-layer walk follows least, C-PR leads that walk: it ranks the transfer family 0.441 above EndorseRank ($[0.423, 0.457]$) and the allowance family 0.206 above AWP ($[0.188, 0.224]$). A node with out-edges in only one layer follows that layer whatever λ is (Equation (3.9)); λ acts only at the nodes with out-edges in both layers, and at T_{obs} only $9{,}459$ contracts are owners in the allowance layer.

On every family S-PR falls below the layer that leads that family. Restarting the transfer walk from the allowance walk lowers the transfer coefficient to 0.782 ($[0.773, 0.791]$; $\Delta\tau = -0.022$, $[-0.030, -0.013]$, against C-PR ($\lambda = 0$)) and raises the allowance coefficient only as far as 0.371 ($[0.350, 0.391]$; $\Delta\tau = -0.467$, $[-0.488, -0.446]$, against C-PR ($\lambda = 1$)). A restart vector changes where the walk lands after each teleport, and it carries only part of the allowance signal into the transfer walk.

As fixed in the analysis plan, the secondary part of rule A asks C-PR's point estimate to lie within the 95% interval of the better layer on the transfer and allowance families, and C-PR to exceed both layers on the stability family. At $\lambda = 0.5$, C-PR fails all three parts, as in the registered analysis (Appendix 8.6). Its transfer coefficient (0.738) lies below the interval $[0.799, 0.809]$ of C-PR ($\lambda = 0$), its allowance coefficient (0.433) below the interval $[0.827, 0.850]$ of C-PR ($\lambda = 1$), and on the stability family it exceeds the allowance layer ($\Delta\tau = +0.352$, $[0.340, 0.365]$) and not the transfer layer (-0.063, $[-0.071, -0.053]$). The paired contrasts of Table 4.19, reported alongside the rule as a stricter check, point the same way: the mixture gives up part of each layer's agreement with its own family and exceeds the better layer on no family.

4.8.2 Rule A in the window W0

Rule A is not met out of window either. The rule asked C-PR to do no worse than C-PR ($\lambda = 1$) on new approval pairs and to beat C-PR ($\lambda = 0$) on new transfer senders in W0 (first block of Table 4.16). At $\lambda = 0.5$, C-PR lies below the allowance layer on new approval pairs ($\Delta\tau = -0.025$, interval $[-0.045, -0.004]$), so the first condition fails, and below the transfer layer on new transfer senders ($\Delta\tau = -0.014$, interval $[-0.019, -0.008]$), so the second fails as well. In the registered analysis the rule was not met either: its primary part failed through its second condition alone, and its secondary part failed (Table 8.16).

Three comparisons outside the rule qualify the first failure. Against its transfer layer walked alone C-PR ranks new approval pairs higher, 0.142 ($[0.124, 0.162]$) against 0.070 ($[0.052, 0.091]$), a paired difference of $+0.071$ ($[0.059, 0.084]$); this contrast, computed after the labels were known, is the

one that the registration of W1 made its criterion F1. Inside the union graph the contrast with the allowance layer does not exclude zero ($+0.020$, $[-0.002, 0.043]$; Table 4.23): on the whole cohort the contrast also reflects that C-PR gives a positive score to the $5{,}700$ spenders of the union graph where its allowance layer reaches $1{,}901$. And against the uncapped allowance layer walked alone C-PR ranks new approval pairs higher ($+0.064$; Section 4.5.5). None of the three decides the rule. The second failure does not depend on these choices: C-PR lies below its transfer layer on new senders inside the union graph (-0.016, $[-0.025, -0.007]$) and with the uncapped layer (-0.019, $[-0.027, -0.012]$). Across the three values of λ the coefficients of C-PR differ by at most 0.005 on either label in either window (Tables 4.15 and 4.20), and λ was not tuned.

The seeded ablation shows where the allowance signal has to sit. S-PR is level with the transfer layer on new senders ($\Delta\tau = -0.003$, interval $[-0.013, 0.008]$) and falls below the allowance layer on new approval pairs ($\Delta\tau = -0.077$, $[-0.098, -0.055]$), which it ranks at 0.091 ($[0.074, 0.113]$) against 0.070 for the transfer layer; C-PR ranks them 0.051 above S-PR ($[0.038, 0.064]$). When transfer edges decide where the walk goes, a restart distribution derived from allowances changes little. The allowance signal reaches the ranking when allowance edges carry the walk.

4.8.3 Rule B in the registered window W1

The registration of Section 3.8.1 was committed to the authors' Git repository after the W0 results of the registered analysis were known and before the study extracted the July–September logs. It fixed rule B with the margin $\delta = 0.01$, the half-width of the W0 interval of the F2 contrast in the registered analysis (0.0040), rounded up. Of rule B the thesis applies two of its registered criteria, F1 and F2; under the registration the rule holds only if each of its criteria holds. F1 asks C-PR to rank new approval pairs above its transfer layer walked alone, with an interval above zero, and F2 asks it not to fall below that layer on new transfer senders by more than δ. Both are applied to the scores of the contract graph as a re-analysis, beside F6, the primary part of rule A on this window, and the spender parts of the two registered sensitivity analyses. Table 4.20 gives every score on the two labels of the spender cohort of Section 4.7.1, and Table 4.21 the contrasts with their verdicts.

Table 4.20: Window W1: Kendall τ between scores frozen at 30 June 2026 on the contract graph and the two spender labels, counted from 1 July 2026 to 30 September 2026. The spenders are the $n = 7{,}915$ contracts among the 7,995 spenders of the labels (79 user accounts and 1 system address are left out); 1,491 of them gained a new approval pair and 1,369 a new sender, and 5,759 (72.8%) lie in the union graph. C-PR ($\lambda = 0$) is the comparator of the two applied criteria of rule B and AWP that of the first sensitivity analysis (Table 4.21); EndorseRank is in Table 4.18. The baselines are raw counts on the contract graph at the freeze. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Score at 30 June 2026	New approval pairs (τ [95% CI])	New transfer senders (τ [95% CI])
In-approve degree at the freeze (baseline)	0.181 [0.158, 0.208]	0.072 [0.046, 0.096]
In-approve value at the freeze (baseline)	0.163 [0.141, 0.188]	0.062 [0.037, 0.084]
Transfer in-degree at the freeze (baseline)	0.031 [0.008, 0.052]	0.302 [0.278, 0.323]
Union in-degree at the freeze (baseline)	0.093 [0.071, 0.114]	0.292 [0.270, 0.312]
AWP	0.064 [0.042, 0.084]	0.314 [0.295, 0.332]
C-PR ($\lambda = 1$)	0.147 [0.124, 0.170]	0.064 [0.041, 0.084]
C-PR ($\lambda = 0$)	0.030 [0.008, 0.050]	0.253 [0.232, 0.272]
C-PR ($\lambda = 0.25$)	0.103 [0.081, 0.124]	0.239 [0.219, 0.258]
C-PR ($\lambda = 0.5$)	0.104 [0.083, 0.126]	0.236 [0.217, 0.255]
C-PR ($\lambda = 0.75$)	0.104 [0.082, 0.126]	0.234 [0.215, 0.252]
S-PR	0.055 [0.034, 0.073]	0.242 [0.222, 0.261]

Rule B is not met. F1 holds: C-PR ranks the spenders that later gained new approval pairs at $\tau = 0.104$ ($[0.083, 0.126]$) against 0.030 ($[0.008, 0.050]$) for its transfer layer walked alone, a difference of $+0.074$ (interval $[0.060, 0.087]$), so the W0 gain recurs. F2 fails. On new transfer senders C-PR reaches 0.236 ($[0.217, 0.255]$) against 0.253 ($[0.232, 0.272]$); the difference is -0.016 ($[-0.022, -0.012]$), and the whole interval lies below the margin of -0.01. The thesis therefore makes no claim that C-PR improves on a walk over the transfers alone. The registered analysis gave the same reading of both criteria (Table 8.16). F1 shows a gain over the transfer layer and not over the approval count: the in-approve degree at the freeze ranks new approval pairs at 0.181, and C-PR lies below it in both windows (Section 4.8.4).

The results reported beside the two criteria agree. F6 fails in both of its parts: C-PR lies below its allowance layer on new approval pairs (-0.043, $[-0.061, -0.024]$) and below its transfer layer on new senders. With AWP as the comparator F1 still holds and F2 fails by more ($+0.040$ and -0.077; Section 4.8.4). With the contracts outside a graph kept as isolated nodes every verdict stays as it was ($+0.073$, $[0.059, 0.086]$, and -0.016, $[-0.022, -0.011]$). Inside the union graph F1's contrast

Table 4.21: Window W1: the paired contrasts F1 and F2, two of the registered criteria of rule B, with their verdicts on the scores of the contract graph, and the spender parts of the two registered sensitivity analyses. Under the registration the rule holds only if each of its criteria holds; this analysis applies F1 and F2 only, so its reading is "not met" when one of them fails and "both applied criteria hold" when both hold. The criteria and the margin are those of the registration of W1; they are applied here as a re-analysis after the registered results were known. $\Delta\tau = \tau(a) - \tau(b)$, with C-PR ($\lambda = 0.5$) as a. Superiority holds when the interval lies above zero; non-inferiority holds when its lower end lies above $-\delta = -0.01$; F6a, the first part of rule A, holds when the interval contains zero or lies above it. Every contrast is on the spenders ($n = 7{,}915$ contracts) of Table 4.20. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	Contrast	Label	Test	$\tau(a)$	$\tau(b)$	$\Delta\tau$ [95% CI]	Verdict
F1 and F2, two of rule B's registered criteria: C-PR ($\lambda = 0.5$) against its transfer layer walked alone, C-PR ($\lambda = 0$)							
F1	C-PR vs. C-PR ($\lambda = 0$)	new approval pairs	superiority	0.104	0.030	+0.074 [0.060, 0.087]	holds
F2	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	non-inferiority, -0.01	0.236	0.253	-0.016 [-0.022, -0.012]	fails
Reading of F1 and F2							not met
Reported beside them							
F6a	C-PR vs. C-PR ($\lambda = 1$)	new approval pairs	no worse	0.104	0.147	-0.043 [-0.061, -0.024]	fails
F6b	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	superiority	0.236	0.253	-0.016 [-0.022, -0.012]	fails
Sensitivity analysis 1: AWP as the comparator							
F1	C-PR vs. AWP	new approval pairs	superiority	0.104	0.064	+0.040 [0.026, 0.055]	holds
F2	C-PR vs. AWP	new transfer senders	non-inferiority, -0.01	0.236	0.314	-0.077 [-0.086, -0.069]	fails
Reading of F1 and F2							not met
Sensitivity analysis 2: contracts outside a graph kept as isolated nodes							
F1	C-PR vs. C-PR ($\lambda = 0$)	new approval pairs	superiority	0.111	0.038	+0.073 [0.059, 0.086]	holds
F2	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	non-inferiority, -0.01	0.238	0.253	-0.016 [-0.022, -0.011]	fails
F6a	C-PR vs. C-PR ($\lambda = 1$)	new approval pairs	no worse	0.111	0.173	-0.062 [-0.081, -0.041]	fails
F6b	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	superiority	0.238	0.253	-0.016 [-0.022, -0.011]	fails
Reading of F1 and F2							not met

is larger ($+0.118$, $[0.101, 0.138]$) and F2's about the same (-0.020, $[-0.027, -0.011]$), and the contrast with the allowance layer does not exclude zero ($+0.007$, $[-0.017, 0.029]$; Table 4.23).

4.8.4 C-PR beside every other score

Table 4.22 sets C-PR against every other score of the study and the four raw counts, on the spender labels of both windows, and Table 4.23 inside the union graph. Only the cells marked in the first table belong to a rule. The contrasts with the union in-degree belong to the definition of the contract-graph analysis, which was made after the registered results, and so the labels, were known; like every other contrast of the table outside the rules, they decide no rule.

Against EndorseRank, the walk over allowances alone, C-PR ranks new approval pairs lower in

Table 4.22: Paired contrasts of C-PR ($\lambda = 0.5$) with every other score of this study and with the four raw counts of the contract graph, on the spender labels of W0 ($n = 7{,}864$ contracts) and W1 ($n = 7{,}915$). Each cell is $\Delta\tau = \tau_{\text{C-PR}} - \tau_b$ over its 95% interval; within a cohort every cell uses the same resamples. Cells marked ^{A} are the contrasts of rule A, fixed in the analysis plan; ^{B} marks F1, F2 and F6a of the registration of W1 and ^{S} the contrasts of its first sensitivity analysis, with AWP as the comparator. These rules were fixed for the raw-unit analysis and are applied here to the scores of the contract graph. Cells marked ^{U}, against the union in-degree, are part of the definition of the contract-graph analysis, defined after the registered results were known and so with the labels known. Like every cell without a mark, also computed after the labels were known, they belong to no rule. The lower block repeats the contrasts with AWP and the in-approve degree on the spenders that had received a transfer from another contract before the freeze, the spenders AWP can order; that restriction was also chosen after the results were known. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	W0 spenders		W1 spenders	
C-PR minus	New approval pairs	New transfer senders	New approval pairs	New transfer senders
C-PR ($\lambda = 1$)	-0.025^{A} [-0.045, -0.004]	+0.170 [0.148, 0.191]	-0.043^{B} [-0.061, -0.024]	+0.173 [0.152, 0.194]
C-PR ($\lambda = 0$)	+0.071 [0.059, 0.084]	-0.014^{A} [-0.019, -0.008]	+0.074^{B} [0.060, 0.087]	-0.016^{B} [-0.022, -0.012]
EndorseRank	-0.026 [-0.046, -0.004]	+0.170 [0.148, 0.191]	-0.043 [-0.062, -0.025]	+0.172 [0.152, 0.193]
EndorseRank (activity restarts)	-0.022 [-0.042, -0.000]	+0.164 [0.142, 0.187]	-0.044 [-0.063, -0.025]	+0.164 [0.144, 0.186]
AWP	+0.044 [0.029, 0.058]	-0.077 [-0.087, -0.068]	+0.040^{S} [0.026, 0.055]	-0.077^{S} [-0.086, -0.069]
S-PR	+0.051 [0.038, 0.064]	-0.011 [-0.021, -0.000]	+0.049 [0.036, 0.063]	-0.006 [-0.015, 0.002]
In-approve degree at the freeze	-0.062 [-0.082, -0.037]	+0.163 [0.142, 0.187]	-0.077 [-0.097, -0.057]	+0.164 [0.141, 0.188]
In-approve value at the freeze	-0.048 [-0.068, -0.024]	+0.173 [0.152, 0.195]	-0.059 [-0.079, -0.039]	+0.174 [0.151, 0.197]
Transfer in-degree at the freeze	+0.072 [0.059, 0.086]	-0.064 [-0.075, -0.054]	+0.072 [0.058, 0.087]	-0.066 [-0.075, -0.057]
Union in-degree at the freeze	+0.016^{U} [0.007, 0.024]	-0.053^{U} [-0.061, -0.043]	+0.011^{U} [0.003, 0.019]	-0.056^{U} [-0.065, -0.047]
Spenders that had received a transfer from another contract before the freeze (W0 $n = 4{,}569$, W1 $n = 4{,}642$)				
AWP	+0.026 [0.013, 0.041]	-0.084 [-0.097, -0.069]	+0.007 [-0.008, 0.021]	-0.094 [-0.108, -0.079]
In-approve degree at the freeze	+0.051 [0.025, 0.076]	—	+0.053 [0.025, 0.080]	—

Table 4.23: The main out-of-window contrasts on the whole spender cohorts and on their members inside the union graph only (W0: 5,700 of 7,864 contracts; W1: 5,759 of 7,915). A spender outside the union graph ends no pair between contracts and scores zero under every score and baseline, so on the whole cohort a contrast also reflects how the two scores order the members with a pair against those without one. The right column decides no rule. $\Delta\tau$ with 95% paired bootstrap interval (400 resamples, seed 42).

Window	Contrast (a minus b)	Label	Whole cohort	Inside the union graph
W0	C-PR minus C-PR ($\lambda = 1$)	new approval pairs	-0.025 [-0.045, -0.004]	$+0.020$ [-0.002, 0.043]
W0	C-PR minus C-PR ($\lambda = 0$)	new transfer senders	-0.014 [-0.019, -0.008]	-0.016 [-0.025, -0.007]
W0	C-PR minus union in-degree	new approval pairs	$+0.016$ [0.007, 0.024]	$+0.035$ [0.022, 0.049]
W0	C-PR minus union in-degree	new transfer senders	-0.053 [-0.061, -0.043]	-0.089 [-0.103, -0.075]
W1	C-PR minus C-PR ($\lambda = 0$)	new approval pairs	$+0.074$ [0.060, 0.087]	$+0.118$ [0.101, 0.138]
W1	C-PR minus C-PR ($\lambda = 0$)	new transfer senders	-0.016 [-0.022, -0.012]	-0.020 [-0.027, -0.011]
W1	C-PR minus C-PR ($\lambda = 1$)	new approval pairs	-0.043 [-0.061, -0.024]	$+0.007$ [-0.017, 0.029]
W1	C-PR minus union in-degree	new approval pairs	$+0.011$ [0.003, 0.019]	$+0.027$ [0.014, 0.041]
W1	C-PR minus union in-degree	new transfer senders	-0.056 [-0.065, -0.047]	-0.093 [-0.107, -0.077]

both windows (-0.026, $[-0.046, -0.004]$, in W0 and -0.043, $[-0.062, -0.025]$, in W1) and new senders higher ($+0.170$, $[0.148, 0.191]$, and $+0.172$, $[0.152, 0.193]$). The variant of EndorseRank with activity restarts gives the same picture: C-PR lies below it on new approval pairs (-0.022, $[-0.042, -0.0004]$, in W0, the upper end just below zero, and -0.044, $[-0.063, -0.025]$, in W1) and above it on new senders.

Against AWP, the walk over transfers alone, the signs are reversed: C-PR ranks new approval pairs higher ($+0.044$, $[0.029, 0.058]$, and $+0.040$, $[0.026, 0.055]$) and new senders lower (-0.077, $[-0.087, -0.068]$, and -0.077, $[-0.086, -0.069]$). Among the spenders that had received a transfer from another contract before the freeze its lead on approvals shrinks to $+0.026$ ($[0.013, 0.041]$) in W0 and is not resolved in W1 ($+0.007$, $[-0.008, 0.021]$), while AWP ranks new senders higher by more (-0.084, $[-0.097, -0.069]$, and -0.094, $[-0.108, -0.079]$) (lower block of Table 4.22; this restriction, too, was chosen after the results were known). On that subset C-PR ranks new approval pairs above the in-approve degree ($+0.051$, $[0.025, 0.076]$, and $+0.053$, $[0.025, 0.080]$).

Against the counts, C-PR does not rank either label above the count of the same edge type. On new approval pairs it lies below the in-approve degree in both windows (-0.062, $[-0.082, -0.037]$, and -0.077, $[-0.097, -0.057]$) and below the in-approve value; on new senders it lies below

the transfer in-degree (-0.064, $[-0.075, -0.054]$, and -0.066, $[-0.075, -0.057]$). It lies above each of these counts on the other label. Against the union in-degree, the count of the edges that C-PR walks, the result is split (Section 4.7), with the same directions and larger gaps inside the union graph: $+0.035$ ($[0.022, 0.049]$) and $+0.027$ ($[0.014, 0.041]$) on new approval pairs, -0.089 ($[-0.103, -0.075]$) and -0.093 ($[-0.107, -0.077]$) on new senders. Against its ablation S-PR it ranks new approval pairs higher ($+0.051$, $[0.038, 0.064]$, and $+0.049$, $[0.036, 0.063]$); on new senders it is slightly lower in W0 (-0.011, $[-0.021, -0.0004]$, the upper end just below zero) and level in W1 (-0.006, $[-0.015, 0.002]$). C-PR thus stands between the two sides on each label, and no score or count of this study has the highest coefficient on both: the in-approve degree has it on new approval pairs and AWP on new senders.

4.9 Claims (C1–C6)

The chapter's results support the claims below, cross-referenced to the hypothesis and research question columns of Table 1.2. C1 rests on the discovery-order test, which was defined after the registered results were known and decides no rule of the analysis plan or the registration. The verdicts of rules A and B in C3 are a re-analysis on the scores of the contract graph; the registered analysis gave the same readings (Appendix 8.6). C2 and C4 rest on contrasts outside every rule. The direct comparisons between EndorseRank and AWP are in Appendix 8.5 and carry no claim of the thesis.

C1. In the run that decides, at T_{obs} in the primary mode, a crawl ordered by C-PR on the contracts read so far reached the contracts with the most user-account demand sooner than a breadth-first crawl ($D_1 = +0.0520$, $[+0.0485, +0.0558]$, on I_T and $+0.0061$, $[+0.0014, +0.0111]$, on I_A) but not sooner than a crawl ordered by the number of contract backlinks, which was sooner ($D_2 = -0.0051$, $[-0.0087, -0.0010]$, and -0.0571, $[-0.0621, -0.0519]$). The verdict is V2 on both importances, so the answer to RQ2 is no (Table 4.12). AWP′ has the highest $\text{AUC}_{10\%}$ of the crawls in that run, apart from the diagnostic δ', every crawl stays far from the

ideal order, and the allowance layer adds a small, resolved amount to C-PR′ over its transfer layer (Tables 4.13 and 4.14). The lead over breadth-first order on I_A does not hold in every sensitivity. In the `out` mode, a sensitivity without a verdict, all four contrasts lie above zero at both freezes, so the result depends on what a visit reveals.

C2. Out of window EndorseRank does not beat the count it smooths, and AWP beats its count by a small margin that does not hold among the contracts with inflow. EndorseRank ranks new approval pairs below the in-approve degree in both windows (W0 -0.036, $[-0.046, -0.027]$; W1 -0.034, $[-0.043, -0.025]$), and so does its variant with activity restarts. AWP ranks new transfer senders slightly above the transfer in-degree on the whole cohorts (W0 $+0.012$, $[0.003, 0.023]$; W1 $+0.011$, $[0.001, 0.022]$) and below it among the W0 spenders that had received a transfer from another contract (-0.035, $[-0.049, -0.022]$) (Tables 4.16, 4.18 and 4.17). Against the union in-degree, the count of the edges it walks, C-PR ranks new approval pairs higher and new senders lower in both windows (Table 4.22). The labels count mostly user accounts, which no score reads, and about 27% of each spender cohort lies outside the union graph and scores zero.

C3. Under its rules, C-PR, a fixed-λ mixture of two dollar-weighted layers with uniform restarts, does not beat its two layers walked alone. Rule A is not met: in W0 C-PR lies below its allowance layer on new approval pairs (-0.025, $[-0.045, -0.004]$) and below its transfer layer on new senders (-0.014, $[-0.019, -0.008]$), and its same-window part fails as well (Tables 4.16 and 4.19). Rule B is not met by its two applied criteria (Table 4.21): F1 holds ($+0.074$, $[0.060, 0.087]$), so the gain over the transfer layer on new approval pairs recurs in W1, and F2 fails (-0.016, $[-0.022, -0.012]$, against the margin of -0.01). Inside the union graph and with the uncapped allowance layer the contrast with the allowance layer reads differently and the contrast with the transfer layer does not (Tables 4.23 and 4.11); neither decides a rule.

C4. Beside the rules, C-PR stands between the two sides on each label in both windows (Table 4.22). Against EndorseRank it ranks new approval pairs lower and new senders higher

(in W0 -0.026, $[-0.046, -0.004]$, and $+0.170$, $[0.148, 0.191]$); against AWP it ranks new approval pairs higher and new senders lower (in W0 $+0.044$, $[0.029, 0.058]$, and -0.077, $[-0.087, -0.068]$). It ranks new approval pairs above S-PR and below the in-approve degree and value, and new senders below the transfer in-degree.

C5. In the same window each single-layer walk agrees most with the proxies of the edge type it walks on, EndorseRank with the allowance family (0.839, $[0.828, 0.849]$) and AWP with the transfer family (0.755, $[0.748, 0.762]$), and the mixtures of C-PR lie between their two layers (Table 4.5). These are intended-construct checks, and part of the agreement is built in. The two walks order the cohort differently: their agreement is $\tau_b = 0.275$ ($[0.258, 0.294]$), only 32.3% of the contract pairs are ordered by both scores, because $5{,}200$ of the $6{,}443$ cohort contracts score zero under EndorseRank, and 26.7% of the pairs that both order are discordant. With the contracts outside a graph kept as isolated nodes EndorseRank's allowance coefficient rises to 0.956 ($[0.952, 0.960]$), and AWP's coefficients do not change (Table 4.6, Figure 4.4).

C6. The cost of each score's solver call grows with the size of the layers it walks and with the iterations its walk needs. The allowance graph has 4.5% of the transfer graph's edges, and AWP takes 27.9 times as long as EndorseRank at an edge ratio of 22.4 (Table 4.3). On the subsamples the time ratio is 0.99 to 1.36 times the edge ratio, and the rows do not test edge counts beyond those of the contract graph (Table 4.4). The low cost belongs to the smaller graph and is no advantage of the algorithm: C-PR walks both layers and takes 1.09 times as long as AWP.